

Segmentation in Pharmaceutical Markets: Evidence from Contraceptives*

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September 24, 2026

Abstract

Many drugs within the same therapeutic class are available both over the counter (OTC) and through health insurance. This paper shows that segmentation of this kind dampens pharmaceutical demand by exposing consumers to hassle costs. We study the hormonal contraceptive market in Colombia, where a price ceiling policy generates variation that separately identifies price sensitivity, preferences for quality, and information frictions about quality. Using a structural model, we find that moving to an OTC-only market would raise contraceptive demand by 20% while modestly reducing retail prices. Moving to an insurance-only market would instead reduce demand by 50% and raise wholesale prices by 2%. These results imply that the hassle costs of obtaining drugs through insurers are the main deterrent to contraceptive use.

Keywords: Market segmentation, Contraception, Information frictions, Price ceilings.

JEL codes: I18, I12, L10.

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1 Introduction

Many drugs within the same therapeutic class are available both over the counter (OTC) and through health insurance. Examples include contraceptives, antihistamines, expectorants, and pain killers. The welfare effects of this form of market segmentation are theoretically ambiguous. On the one hand, it can increase product variety and consumer choice, raising welfare. On the other, it can soften product competition and raise prices or expose consumers to non-price costs from navigating different segments, lowering welfare. Characterizing these effects is important for understanding whether OTC approval achieves the access gains it is meant to deliver and whether price discrimination facilitated by market segmentation undermines the ability of insurance formularies to contain pharmaceutical spending.¹

We study the effects of segmentation in the hormonal contraceptive market in Colombia. This setting is particularly important because expanded access is socially valuable: access to contraceptives has been shown to significantly improve women’s economic opportunities and well-being (Bailey, 2006; Goldin and Katz, 2002). Contraceptives are available OTC in 56 countries (Grindlay et al., 2013), and the set of drug classes subject to this form of segmentation is actively expanding as more drugs are approved for OTC sale.² We show that market segmentation hurts demand and access primarily because of the non-price costs of navigating the insurance segment. softens product competition and allows manufacturers to price drugs higher than in a pooled market, which hurts demand and access. Moreover, unlike markets where segmentation is common and consumers are well-aware of product attributes (Karle et al., 2020; Marshall, 2015), information frictions about product quality in pharmaceutical markets (Crawford and Shum, 2005), further amplify the effects of segmentation.

The Colombian market is uniquely informative for studying this question. In the

¹Many countries have expanded OTC availability for drug classes where unsupervised use is safe to alleviate health systems from consultations that are not always clinically necessary.

²For example, in December 2024, the U.S. Food and Drug Administration (FDA) finalized the Additional Conditions for Nonprescription Use (ACNU) rule, creating a pathway for drugs that previously required medical supervision to be approved for OTC sale in different countries (FDA, 2024). Under this rule, a progestin-only pill became available OTC for the first time, even as the Affordable Care Act already mandated comprehensive insurance coverage of contraceptives (FDA, 2024).

OTC segment, women pay full retail prices and have access to all government-approved contraceptives. In the insurance segment, women pay nothing out of pocket but face a restricted formulary and hassle costs from interacting with the healthcare system.³ Our analysis combines product-level data on monthly retail prices and sales from IQVIA Colombia for the OTC segment with individual-level data on enrollment and health claims covering nearly half of the Colombian population for the insurance segment, both between 2018 and 2019.

We begin by documenting the consequences of market segmentation in reduced form, exploiting a 2019 policy in which the Colombian government imposed wholesale price ceilings on 42 contraceptives, half of which were marketed as higher-quality products because of their milder side effects. Using an event study design that compares targeted against untargeted products, we find that retail prices fell 48% for products offered exclusively OTC and 63% for products simultaneously available through insurance. The difference in treatment effects reflects the fact that pre-regulation prices for OTC-only products were much higher than their ceilings compared to dual-segment products, suggesting that segmentation allows manufacturers to set relatively high prices when the insurance segment does not offer a comparable alternative.

Replicating the event study exercise in the insurance segment data reveals that wholesale prices for targeted products fell 28% and their demand nearly doubled by the end of the sample period. However, changes in demand in this segment cannot be explained by price reductions as women face zero out-of-pocket prices.

We provide evidence that this demand response reflects mainly a reduction in information frictions. Using a survey of 310 reproductive-age women in Bogotá, we find that although 87.6% of respondents recognize at least one brand, 24.7% cannot correctly identify the route of administration of any brand, indicating that information frictions are widespread. Knowledge is higher among women with any healthcare spending than among those without, further indicating a correlation between care-seeking behavior and information frictions. The claims data corroborate the survey: better-informed women

³Transaction costs, time costs of scheduling doctor appointments, and travel costs to insurer medication-dispensing facilities are examples of hassle costs women face when claiming contraceptives through insurance.

(those with above-median pre-policy healthcare spending) exhibit 40% higher demand for targeted products (i.e., higher-quality products) than less-informed women, and this gap halves after the policy, suggesting that the regulation acted as an information signal that updated the beliefs of less-informed women.

We also find evidence of hassle costs in the insurance segment. These reflect the opportunity cost of scheduling a doctor's appointment to obtain a prescription, waiting at insurer dispensing facilities, and navigating authorization procedures before a claim is approved. Our survey shows that among women who obtain contraceptives through insurance, 82.4% report that obtaining the product takes more than one hour, compared with 79.9% of OTC purchasers who obtain it in under one hour.⁴ In the claims data, measures of hassle costs are also strongly correlated with the wholesale prices insurers pay manufacturers, consistent with insurers imposing greater administrative barriers for more expensive products, and suggesting that wholesale prices serve as a proxy for hassle costs in the structural model.

To quantify the equilibrium effects of segmentation, we develop and estimate a structural model of the contraceptive market. We model demand among women seeking to avoid pregnancy using a random-coefficients logit framework in which women choose among available hormonal contraceptives and an outside option to maximize their indirect utility. Demand depends on contraceptive prices, hassle costs, and beliefs about product quality. Wholesale prices proxy for hassle costs in the insurance segment. Pre-policy healthcare spending, in turn, determines women's information about product quality. We assume women are Bayesian agents who update their beliefs about contraceptive quality upon observing whether a product is regulated (since the regulation targeted higher-quality products). On the supply side, manufacturers sell through both market segments and set segment-specific prices, competing Nash-Bertrand in the OTC segment and making take-it-or-leave-it offers to a representative insurer in the insurance segment.

Our estimates show that women place 20% weight on their prior beliefs about product quality and the remaining 80% on quality signals, implying that most women remain

⁴In Colombia, medications approved by insurers must be claimed at insurer-owned facilities instead of pharmacies as in the United States, and wait times at these facilities are relatively long (see [El Colombiano, Dec. 21, 2023](#)).

poorly informed in the absence of a policy shock. The median own-price elasticity in the OTC segment equals -1.73 . Price sensitivity is higher among women who are younger, lower income, and more likely to have claimed contraceptives through insurance. These estimates are consistent with those from the randomized controlled trial in [Bailey et al. \(2025\)](#), which studies how out-of-pocket cost subsidies affect contraceptive method choice among uninsured women in the U.S. Women also face substantial hassle costs when obtaining contraceptives through insurance. They are willing to pay \$8 on average to avoid these hassle costs, but retail prices in the observed market exceed that threshold preventing switches to the OTC market. On the supply side, we find that the average markup equals 62% of the retail price in the OTC segment and 24% of the wholesale price in the insurance segment.

Using the model estimates, we simulate counterfactual scenarios that eliminate segmentation by moving to an OTC-only or an insurance-only market. The OTC-only scenario can be interpreted as granting non-prescription status to all contraceptives, while the insurance-only scenario corresponds to requiring the national insurance formulary to cover all products. Additionally, we remove information frictions to separate their contribution to welfare from that of retail prices and hassle costs.

We find that market segmentation dampens contraceptive demand. Moving to an OTC-only market raises demand by 20% relative to baseline, or roughly 936,000 additional women using contraceptives each month, with the largest gains for conventional pills. This response is driven primarily by the removal of the hassle costs women face when obtaining contraceptives through insurance. Equilibrium prices fall only modestly, but the aggregate effect on out-of-pocket spending is substantial: a reduction of 5.2 billion pesos per year. In this scenario, consumer surplus increases by \$3.7 per woman-year, roughly 65% of the average retail price. Moving to an insurance-only market has the opposite effect. Total demand falls by 51% as hassle costs extend to all products, and wholesale prices rise by 3% because the insurer's participation constraint does not offset the higher marginal costs of products previously sold over the counter. Likewise, consumer surplus falls by \$8.7 per woman-year.

Our paper makes three contributions to the literature. First, the literature on market

segmentation has mostly studied markets characterized by adverse selection ([Shahidinejad et al., 2025](#)), such as health insurance, and found that pooling the market results in lower premiums ([Fleitas et al., 2019](#); [Dickstein et al., 2024](#)). We contribute by showing that the effects of pooling are similar in markets without selection but where segments differ in prices and hassle costs.

Second, the industrial organization literature studying information frictions has mostly focused on its impact on product demand ([Brown et al., 2023](#); [Dickstein et al., 2023](#); [Brown and Jeon, 2024](#); [Gupta and Cao, 2025](#)). Within pharmaceutical markets, the literature has mainly studied information frictions as an explanation for why consumers buy branded drugs over generics ([Handel and Schwartzstein, 2018](#); [Bronnenberg et al., 2015](#)). Our main contribution is to jointly model demand and supply (as in [Atal et al., 2024](#)), considering how information frictions interact with segmentation in equilibrium. Our results also suggest that providing information about contraceptive quality can reduce manufacturer's ability to exert market power.

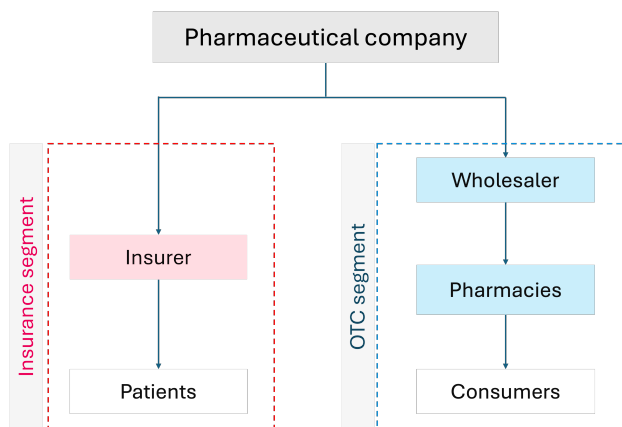
Third, while a rich literature in health economics has studied demand-side barriers to contraceptive use, ranging from financial constraints ([Willage, 2020](#); [Bailey, 2012](#); [Pace et al., 2016](#); [Snyder et al., 2018](#); [Bailey et al., 2023](#); [Dupas et al., 2024](#)) to cultural and social barriers ([Miller et al., 2023](#); [Bau et al., 2024](#)), this work has focused almost exclusively on insurance markets or used randomized controlled trials in low-income settings. Direct evidence from OTC markets remains scarce despite their prevalence across countries, and the supply-side role of market structure in shaping contraceptive access has received little attention ([Rossin-Slater et al., 2026](#)). We fill this gap by providing the first evidence on how segmentation between OTC and insurance markets affects women's access to contraception, with direct implications for the regulation of OTC contraceptive approvals.

2 Setting

2.1 Institutional Background

We study the welfare consequences of market segmentation in the Colombian hormonal contraceptive market, where products are sold through both OTC and insurance segments. The OTC segment accounts for roughly 70% of contraceptive sales by volume, and the insurance segment makes up the remaining 30% (Appendix Figure 1). Contraceptives in Colombia are known as “branded generics,” meaning that products are off-patent, yet are sold under distinct brand names rather than under their chemical designation. Segmentation between OTC and insurance therefore does not reflect a distinction between branded and generic drugs.

FIGURE 1: Pharmaceutical Supply Chain in Colombia



Insurance segment. In the insurance segment, health insurers negotiate wholesale prices with pharmaceutical manufacturers or distributors for contraceptives covered by the national health insurance plan (see Figure 1). The set of contraceptives covered by the health plan is determined by the government on the basis of clinical effectiveness, and is therefore exogenous to individual insurers. Approximately 60 brand-dosage combinations (hereafter, “products”) are covered by the health plan, including first- and second-generation oral contraceptives, implants, intrauterine devices (IUDs), and injections.

First- and second-generation oral contraceptives contain older progestin formulations associated with higher rates of side effects such as weight gain, acne, and irregular bleed-

ing, whereas third- and fourth-generation formulations use newer progestins with milder androgenic side effects (Sitruk-Ware, 2006). Notably, third- and fourth-generation oral contraceptives are not covered by the national health insurance plan. When women obtain a covered contraceptive, insurers pay the negotiated wholesale price to the manufacturer or distributor, but women face zero out-of-pocket costs.

Health insurance coverage in Colombia is nearly universal: roughly 93% of Colombians are enrolled (DANE, 2019), so selection into insurance is limited. All insurers are required by law to cover the same set of contraceptives, so women face the same formulary regardless of which insurer they belong to. The decision of which insurer to enroll with is therefore orthogonal to contraceptive demand, and the relevant selection margin is whether women choose to obtain contraceptives OTC or through their insurer.

Accessing contraceptives through insurance requires a doctor's prescription, which takes time and effort to obtain. Doctors recommend contraceptives based on the woman's health status and family planning needs, which supports treating women as the primary decision-makers.

OTC segment. In the OTC segment, pharmaceutical manufacturers negotiate wholesale prices with distributors, who then supply pharmacies. Pharmacies set the retail prices faced by consumers, though strong competition among pharmacies suggests that pricing power rests primarily with manufacturers rather than retailers.⁵ Women purchasing OTC have access to the full set of government-approved contraceptives (including those offered through insurance), nearly 150 products.

Some OTC products have historically been considerably expensive. For example, before 2019, the most expensive fourth-generation oral pill had a retail price equivalent to 15% of the monthly minimum wage. Although a doctor's prescription is technically required for pharmacy purchases, enforcement is minimal, making it a *de facto* OTC market.⁶ Women purchasing OTC therefore face high prices but few hassle costs, while women sourcing

⁵See for example https://www.bloomberglinea.com/latinoamerica/colombia/el-auge-de-los-genericos-la-estrategia-de-ahorro-que-transforma-el-mercado-farmaceutico/?reader_id=READER_ID&url=SOURCE_URL&_=RANDOM/

⁶An audit study in Bogotá found that only 1% of pharmacy staff requested a prescription when selling contraceptive pills (Andia et al., 2022).

them from insurance face zero out-of-pocket prices but substantial hassle costs.

Based on the structure of the market, women must first select a market segment and then a product within that segment. This choice is inherently complex. It requires trading off prices and hassle costs for segment choice and then learning about contraceptive quality through experimentation for product choice, but experimentation is costly because switching products can result in unintended pregnancy. Hence, women often face information frictions about quality when choosing their contraceptive.

2.2 Price Ceiling Policy

To facilitate contraceptive choice and reduce out-of-pocket costs for women purchasing OTC, the Colombian government imposed wholesale price ceilings on 29 contraceptives in January 2019. Half of the targeted products were third- and fourth-generation oral contraceptives—commonly marketed as higher quality—whose prices prior to the policy were nearly three times higher than in countries such as Canada and Norway ([Ministerio de Salud y Protección Social, 2019](#)). The regulation was subsequently expanded in July 2019 and May 2020 to cover 13 additional products, bringing the total number of regulated contraceptives to 42.⁷

The contraceptive price ceilings were implemented under a broader pharmaceutical price regulation framework introduced in 2013, which targeted products based on two criteria. First, products had to belong to highly concentrated markets—with either fewer than three competitors or a Herfindahl-Hirschman Index exceeding 2,500—and have the highest market share in their market.⁸ Second, their retail prices had to exceed an international reference price, defined as the 25th percentile of average prices for the same active ingredient across a set of comparator countries.⁹

⁷Additional details on regulated products are provided in Circulares 07/2018, 09/2019, and 10/2020 from the Ministry of Health available at [MinSalud, 2025](#).

⁸Markets were defined as the fifth level of the World Health Organization's Anatomical Therapeutic Chemical (ATC) classification (ATC5).

⁹Reference countries include Argentina, Brazil, Chile, Ecuador, Mexico, Panama, Peru, Uruguay, Spain, the United States, the United Kingdom, Australia, Canada, France, Norway, Germany, and Portugal.

3 Data

We use three datasets for our analysis. The first is IQVIA Colombia, which reports monthly retail prices, quantities, national drug codes (NDCs), brand names, manufacturers, and routes of administration for all contraceptives (3-digit ATC G03) sold OTC between 2018 and 2019, covering approximately 90% of Colombian pharmacies. The retail pharmacy market consists of large pharmacy chains with stores in multiple cities, which account for 30% of units sold in the data, and small independent drugstores, which account for the remaining 70%. While pharmacy chains report retail prices directly, independent pharmacies report only wholesale prices. Hence, to approximate retail prices for these observations, we apply a 15% markup following IQVIA’s guidance.

The second dataset are the quarterly adverse event reports from the FDA’s Adverse Event Reporting System (FAERS). We use these reports—which cover unintended pregnancies, venous thromboembolism, weight gain, acne, bleeding, device expulsion, and other outcomes—to construct a measure of contraceptive quality at the level of active ingredient, route of administration, and quarter. For active ingredients approved in Colombia but not in the U.S., we supplement the FAERS data with Colombia’s National Drug and Safety Agency (INVIMA) adverse event reports. Our quality measure is the Proportional Reporting Ratio (PRR) typically used in pharmacovigilance ([Evans et al., 2001](#)). The PRR compares the share of a drug’s adverse event reports attributable to a given reaction relative to the share across all other drugs. Appendix 1 provides details on the construction of this measure.

The third dataset are individual-level enrollment and health claims records from the Colombian Ministry of Health and Social Protection (known as *Base Única de Afiliados* and *Base de Suficiencia*, respectively). These data cover all outpatient, inpatient, and prescription claims made by nearly 25 million enrollees in Colombia’s contributory health insurance system between 2018 and 2019. For each claim, we observe the contraceptive prescribed, and the woman’s insurer, age, and municipality of residence.

We define a product as a brand–dosage combination and aggregate the IQVIA data to the product level. We similarly aggregate individual-level health claims to the product

TABLE 1: Summary Statistics by Market Segment

Variable	Contraceptive type	Pre 2019-01		Post 2019-01	
		mean	sd	mean	sd
<i>Panel A. Over-the-counter</i>					
<u>Targeted</u>					
Prices	Conventional pill	11.44	(7.84)	4.53	(3.56)
	Other	3.66	(1.63)	3.25	(1.53)
Sales volumes	Conventional pill	13.34	(18.97)	18.39	(23.42)
	Other	24.52	(20.05)	35.20	(24.25)
<u>Untargeted</u>					
Prices	Conventional pill	5.33	(5.83)	4.52	(4.77)
	Injection	4.47	(2.54)	4.94	(2.59)
	Other	2.77	(1.64)	2.79	(1.64)
Sales volumes	Conventional pill	7.26	(21.90)	8.19	(24.82)
	Injection	44.98	(32.16)	47.49	(41.45)
	Other	21.22	(33.75)	19.24	(33.97)
Total sales volume		1450.50	(87.67)	1654.09	(84.78)
<i>Panel B. Insurance</i>					
<u>Targeted</u>					
Prices	Conventional pill	3.47	(0.93)	2.30	(0.24)
	Other	1.49	(1.17)	1.66	(0.75)
Sales volumes	Conventional pill	1.91	(3.02)	3.11	(4.17)
	Other	1.74	(2.52)	1.72	(2.21)
<u>Untargeted</u>					
Prices	Conventional pill	0.74	(1.12)	0.75	(1.25)
	Injection	0.81	(0.27)	0.81	(0.28)
	Other	0.67	(0.13)	0.65	(0.02)
Sales volumes	Conventional pill	2.89	(5.97)	2.92	(5.94)
	Injection	20.11	(20.16)	16.62	(22.91)
	Other	20.79	(41.01)	23.13	(42.38)
Total sales volume		289.54	(31.85)	260.06	(62.11)

Note: Table shows the mean and standard deviation in parentheses of prices and sales volumes by contraceptive type, regulation status, and policy period from January 2018 to December 2019. *Targeted* products correspond to those products regulated in January 2019 or July 2019. Products regulated in 2020 are classified as *Untargeted* because they were not regulated during the sample period. Panel A shows summary statistics for the OTC segment and Panel B for the insurance segment. In the insurance segment, we exclude data for implants that were introduced for the first time after January 2019. Average prices are weighted by sales volumes. Prices are measured in 2018 USD, and sales volumes are in 1,000 units.

level, restricting attention to claims made by women of reproductive age (15–45). To ensure comparability across contraceptive presentations, we standardize prices and quantities to a single menstrual cycle. For example, for an NDC corresponding to three blisters of 21 pills, we divide the price by three and multiply the quantity by three. Appendix 1 provides additional details on data cleaning.

Table 1 reports summary statistics for the OTC and insurance segments before and after the first regulatory wave in January 2019. An observation in this table is a product-month. Prior to the regulation, targeted products in the OTC segment had substantially higher prices than untargeted products. For example, the average price of a targeted conventional pill (\$11.44) was more than twice that of an untargeted one (\$5.33). After the policy, prices of targeted products declined to levels comparable to those of untargeted products, and their sales volumes increased. The policy also generated spillover effects, as prices of untargeted products within the same contraceptive type as targeted ones declined after the regulation. Overall, total sales volume in the OTC segment increased 14%. In the insurance segment, wholesale prices for targeted contraceptives also declined after the policy, although the effects were smaller in magnitude. For example, the average wholesale price of a conventional pill declined from \$3.47 to \$2.30. Spillover effects are not apparent in this segment.

4 Reduced-Form Evidence of Market Segmentation

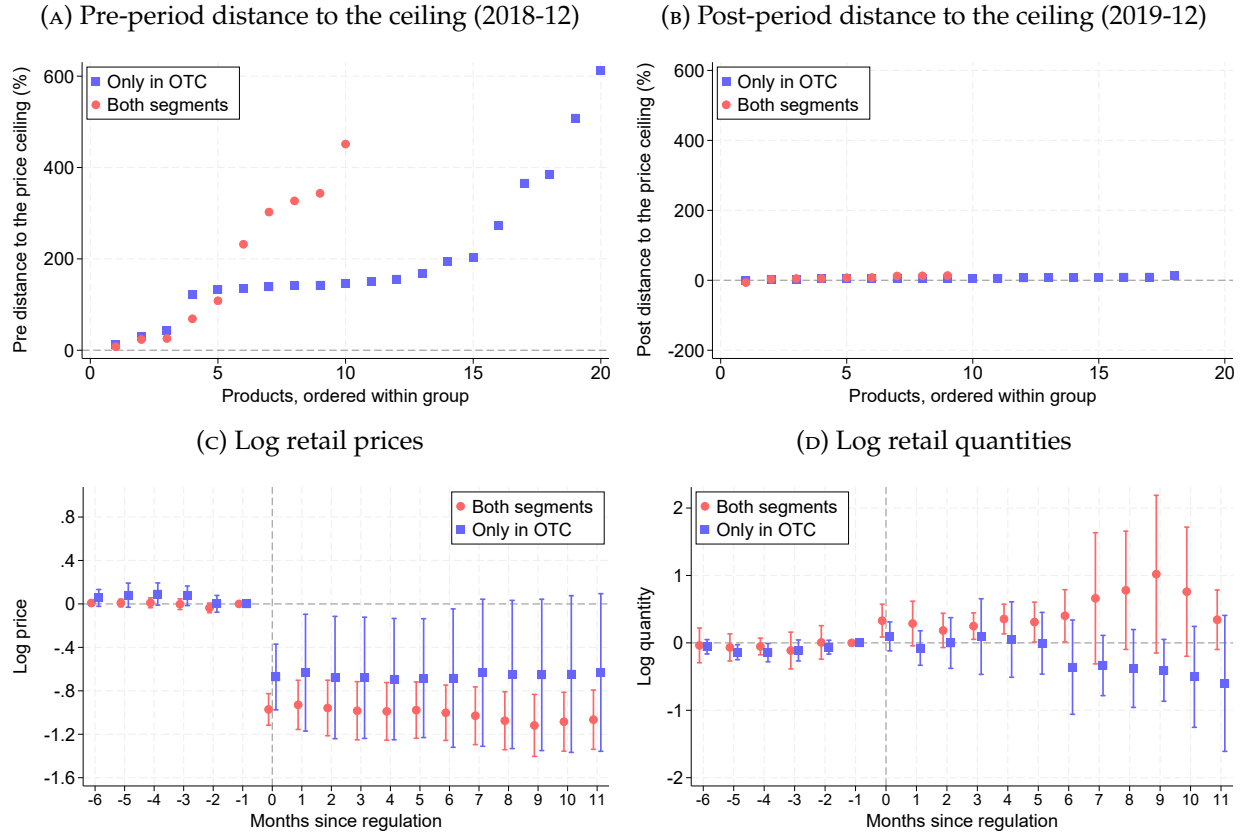
As a first step, we use the price ceiling policy to document the potential equilibrium consequences of market segmentation in reduced form. We estimate an event study on the OTC segment, comparing products targeted by the price ceilings against untargeted products:

$$y_{jt} = \beta_{\tau} + \beta_{\tau}^D \cdot \text{Dual}_j + \delta_j + \lambda_{b(j)t} + \gamma_t + \varepsilon_{jt},$$

where y_{jt} is an outcome for product j in month t , τ is the number of months since the regulation wave in January 2019 (normalizing untargeted products to -1), Dual_j indicates products sold simultaneously in both segments, and δ_j and γ_t are product and month fixed effects, respectively. We include fixed effects $\lambda_{b(j)t}$ for the interaction between a product's average pre-policy price decile and calendar month, so that treatment effects are identified within groups of products priced similarly before the regulation. The coefficients β_{τ} capture the response of OTC-exclusive products, while β_{τ}^D measures the differential response of dual-segment products. Because the price ceilings were set as a function of pre-policy

prices, $\lambda_{b(j)t}$ prevents differences in nominal ceilings from mechanically generating differences in estimated treatment effects across segments. If segmentation shields OTC-only manufacturers from competition, then β_{τ}^D should be negative, as OTC-only products start farther from their ceilings. We focus on a balanced panel of products, use a two-way fixed effects estimator, and use two-way clustering of standard errors at the product and manufacturer level.

FIGURE 2: Heterogeneous Effects of the Price Ceiling Policy in the OTC Segment



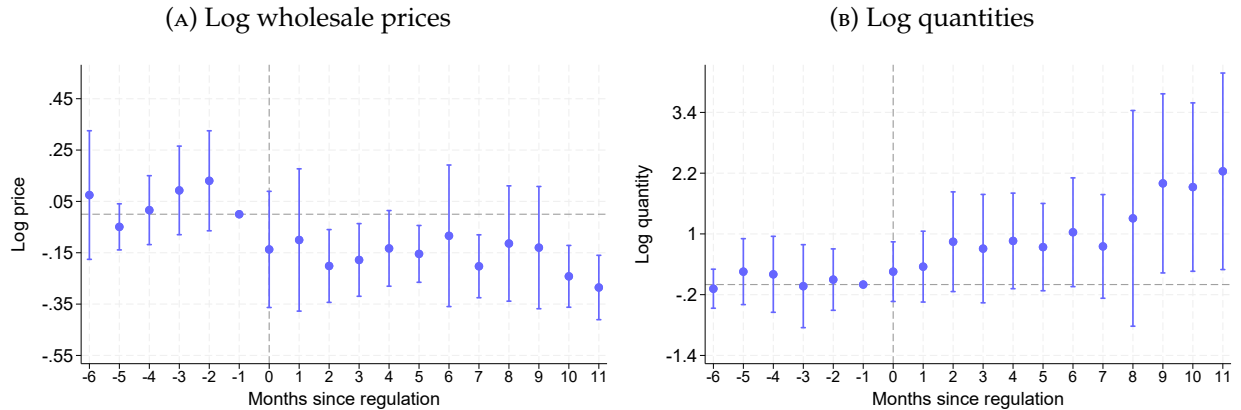
Note: Panel A presents the distance from the retail price to the price ceiling in December 2018 for targeted products. Panel B presents the distance from the retail price to the price ceiling in December 2019 for targeted products. In both panels, products are ordered based on the distance and presented separately for the group offered OTC exclusively (blue) the group offered in both segments (red). Each dot corresponds to one product. Panel C presents event study coefficients and 95% confidence intervals of a TWFE estimator on log retail prices. Panel D reports the analogous estimates for log retail unit sales. An observation in these regressions is a product-month. Estimates in blue report the dynamic treatment effects on OTC-only products. Estimates in red report the dynamic treatment effects on products offered in both segments.

Identification of treatment effects relies on two assumptions: that outcomes for targeted products would have evolved in parallel to untargeted ones absent the regulation (indirectly testable with pre-period estimates), and that the control group is unaffected by the

regulation. Because the regulation generated small spillovers onto untargeted products of the same contraceptive type (Table 1), we expect estimates to be slightly conservative.

Figure 2, Panel A starts by showing that OTC-only products on average were priced much higher than the ceiling (203%) compared to dual-segment products (177%). One year after the policy, prices for all products fall to the price ceiling (Panel B). Panel C presents our estimates for β_τ and $\beta_\tau + \beta_\tau^D$. Retail prices for dual-segment products fell by 63% ($= 100 \times (e^\beta - 1)$) in the month after the regulation, compared with 48% for products offered exclusively OTC. OTC-only products adjusted less presumably because segmentation allowed manufacturers to sustain higher prices for these products at baseline.¹⁰ Panel D shows a consistent pattern in quantities: dual-segment products, whose prices fell more, also experienced the larger demand increase.

FIGURE 3: Impact of Price Ceilings on Equilibrium Prices and Quantities in Insurance Segment



Note: Figure presents event study coefficients and 95% confidence intervals estimated using a TWFE estimator. Panel A shows the effect of the price ceiling policy on log wholesale prices, and Panel B reports the analogous estimates for log unit sales in the insurance segment. An observation is a contraceptive month.

We next estimate the same event study on claims data from the insurance segment, dropping the Dual_j interaction and the $\lambda_{b(j)t}$ fixed effects, since there are too few insurance-only products to identify these parameters. Figure 3 shows that wholesale prices for targeted products fell 28% (Panel A) and that demand for these products more than doubled after the regulation, particularly toward the end of 2019 (Panel B). The demand response within the insurance segment is difficult to attribute to prices, since women face

¹⁰The smaller treatment effect among OTC-only products could also be driven by spillover effects onto untargeted drugs being larger than among dual-segment products. In any case, the stark differences between OTC-only and dual-segment products suggests that segmentation is consequential for market outcomes.

zero out-of-pocket costs to begin with. This points to the existence of additional factors shaping contraceptive demand, such as some women interpreting the policy as a signal of product quality or insurers making it easier for women to access targeted drugs.

5 Survey and Market Evidence on Non-Price Frictions

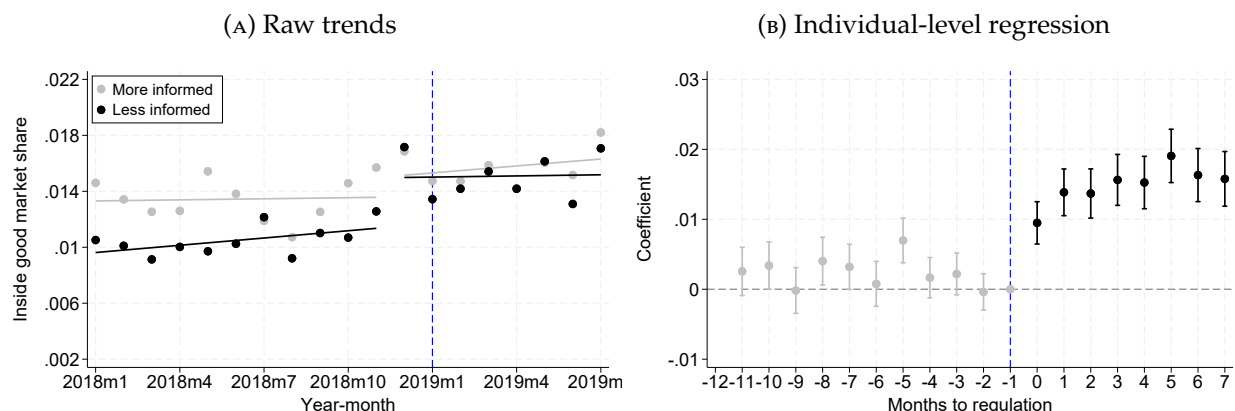
Information frictions. To understand the forces beyond prices that determine demand through each market segment, we surveyed 310 women of reproductive age in Bogotá from June to August 2026. We were interested in measuring women’s awareness of product attributes given that information frictions are a well-known feature of pharmaceutical markets ([Crawford and Shum, 2005](#)), and in identifying the main barriers women perceive from sourcing their contraceptives OTC or through insurance.¹¹ An extract of survey questions and a description of respondents is provided in Appendix 2. 77% of respondents were 18–25 years old, 73% had more than a high school education, and 54% reported currently using contraceptives. On educational attainment and current contraceptive use, survey respondents closely matched contributory-system respondents in the 2015 Demographic and Health Survey (DHS), but reported much lower rates of injection use (Appendix Table 2).

Respondents were presented with seven commonly marketed contraceptive brands spanning oral pills, injections, and implants. For each brand, we first asked whether they recognized its name and then asked them to identify objectively verifiable characteristics, including route of administration. Appendix Table 3 shows that brand awareness is substantially more common than product knowledge. Although 87.6% of respondents recognize at least one brand, 24.7% cannot correctly identify the route of administration of any recognized brand. Familiarity with a brand name therefore does not imply knowledge of readily observable product characteristics nor complex unobservable ones like quality, supporting the existence of information frictions (rather than limited consideration set

¹¹We note that our survey is externally valid: 61.1% obtained their most recent contraceptive from a pharmacy, whereas 13.5% obtained it from an insurer dispensing facility; this figures are comparable with those obtained from administrative records such as the Medication Price Information System (SISMED) from the Ministry of Health and Social Protection.

awareness). Knowledge of product attributes also differs by past healthcare expenditure: 21% of those with any health spending in the past month cannot correctly identify a route compared to 46% of their counterparts.

FIGURE 4: Targeted Product Market Shares Among Informed and Uninformed Women



Note: Panel A presents the trend of average targeted product market share among more- and less-informed women in Colombia's contributory health insurance system. More-informed women are those with above-median pre-policy annual healthcare spending (excluding prescriptions); less-informed women are their counterparts. Panel B presents coefficients and 95% confidence intervals of an event study comparing the likelihood of consuming targeted products between less-informed (treated) and more-informed (control) women. The regression includes a treated dummy and month fixed effects; standard errors are clustered at the woman level. The dashed blue line represents the month when the price ceiling regulation went into effect.

The market data corroborate the survey. Figure 4 compares contraceptive use in the claims data between women with above- and below-median annual healthcare spending (excluding prescriptions), which we classify as better- and less-informed women, respectively, based on the survey findings. Panel A plots the average market share of products targeted by the price ceilings, which are considered of higher quality, in each group. Better-informed women exhibit higher baseline use of targeted products, and less-informed women show a stronger post-policy increase, so the gap between the two groups narrows after the regulation. Panel B formalizes these differences with an event study using individual-level data, comparing the likelihood of claiming a targeted contraceptive between less-informed (treated) and better-informed (control) women. Less-informed women become 1.5 percentage points more likely to claim targeted products after the policy, a 23% increase relative to the baseline treated mean. Appendix Figure 2 shows that effects survive even in a balanced panel of women including individual fixed effects. These patterns suggest that women interpreted the policy as a signal of product quality:

most targeted products are marketed as having milder side effects, so a regulation singling them out conveyed information that less-informed women did not previously hold. Consistent with this interpretation, Appendix Figure 3 shows a sharp increase in Google searches for the active ingredients targeted by the policy around January 2019.

We explore several competing explanations for the observed increase in demand for targeted products in the insurance data in Appendix 3.1, including changes in insurer product inventories and experimentation, noting that the data are inconsistent with those mechanisms.

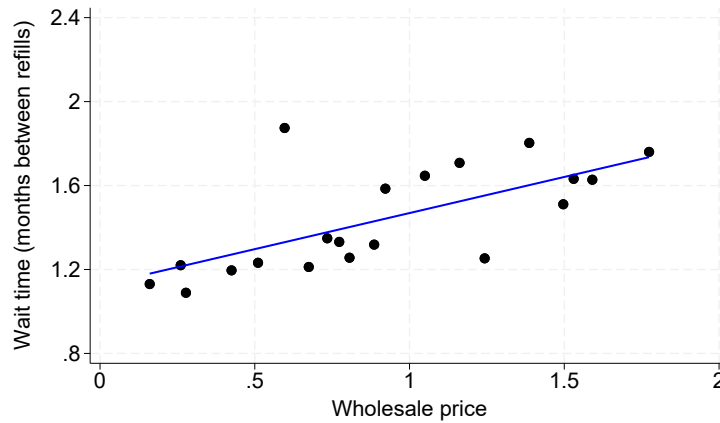
Hassle costs. Information frictions are an important determinant of contraceptive demand. However, because they affect women across both market segments to varying degrees, they cannot alone explain why most sales occur OTC. Another factor that can help explain the dominance of OTC purchases is the hassle associated with obtaining contraceptives through insurance. Purchasing a contraceptive OTC generally requires only a visit to a retail pharmacy, whereas obtaining it through insurance may involve appointments, prior authorizations, or travel to an insurer dispensing facility. These additional costs may discourage women from claiming contraceptives through insurance even when doing so would be financially advantageous.¹² The survey provides direct evidence of this. Appendix Table 4 shows that 77.6% of respondents have a pharmacy within ten minutes, compared with only 5.4% for an insurer dispensing facility. Conversely, 65.2% report traveling more than thirty minutes to an insurer facility, compared with 5.5% to a pharmacy.

Beyond travel costs which impact all products equally, hassle costs may also vary across products. Insurers may have stronger incentives to manage utilization of more expensive medications because they are more consequential for profits. If insurers engage in this type of utilization management, women seeking access to more expensive contraceptives would face greater hassle costs than those seeking cheaper alternatives. To examine this, we construct a measure of hassle costs using the health claims data. For women who consume products within a single route of administration, we define the hassle cost as

¹²In Colombia, long wait times and administrative burdens in medication dispensing are among the most common complaints filed against health insurers (See [El Colombiano](#), Dec. 21, 2023).

the average number of months she has to wait between consecutive refills conditional on the number of dispensed units. For example, if a woman claims a 3-month pack of pills in January but we see her making her next refill in May, then her wait time is one month. Figure 5 presents a binned scatter plot of this wait time against the wholesale price, after residualizing both variables on route of administration fixed effects to ensure that the correlation is identified from within-route variation rather than across routes.¹³ In line with the intuition, contraceptives with higher wholesale prices tend to have longer wait times.

FIGURE 5: Correlation Between Wait Time and Wholesale Price



Note: Figure presents a binned scatter plot of wait time between refills against wholesale prices in the insurance segment. We start with the individual-level health claims data and use contraceptive claims for women aged 15-50 who consume contraceptives of only one route of administration during the sample period. For each woman we construct the number of months between consecutive prescriptions. Then, we average the refill gap and the wholesale price per product and month. Finally, we construct the binned scatter plot on 20 equally-sized bins of the wholesale price, reporting the average refill gap in the bin.

The correlation admits other interpretations as well. For instance, higher-priced contraceptives may require more complex clinical assessments before being prescribed, or women using more expensive contraceptives may have stronger preferences for specific products and therefore are willing to wait longer to obtain them rather than switching to a cheaper alternative. Although we cannot rule out these explanations, Appendix Table 5 shows that at the product-level, conditional on quality, wholesale price is a significant determinant of product market share even if women do not face these prices directly. Given this evidence, in our model we use insurers' wholesale acquisition costs as a proxy

¹³This approach explicitly avoids making comparisons between short- and long-acting methods as well as between contraceptives that are typically prescribed for one versus three months.

for the hassle costs women face when accessing contraceptives through insurance.

6 Model

The reduced-form analyses so far yield four main takeaways: (i) market segmentation can be consequential for prices and demand by affecting the degree of competition, (ii) contraceptive demand responds strongly to OTC retail prices, (iii) women face information frictions about quality and update beliefs based on quality signals related to the price ceiling regulation, and (iv) women face hassle costs when accessing contraceptives through insurance. In this section, we develop a model of the Colombian market for hormonal contraceptives that incorporates these takeaways.

6.1 Demand

Contraceptives are supplied through two segments, indexed by $h = \{ins, otc\}$, where *ins* denotes insurance and *otc* denotes over-the-counter. The set of products available to consumers varies across segments, with products indexed by $j = 1, \dots, J_{ht}$, and $|J_{ht}|$ denoting the cardinality of the set. Assume that, conditional on the set of products available in month t through market segment h , women have common prior beliefs about contraceptive j 's quality given by:

$$q_j | J_{ht} \sim \text{Beta}(\mu, \omega),$$

where mean quality is $E[q_j | J_{ht}] = \mu / (\mu + \omega)$.¹⁴ Women receive a signal every month, denoted r_{jt} , about whether contraceptive j is regulated. The product-specific signal depends on the quality of the drug,¹⁵ with

$$r_{jt} \sim \text{Bernoulli}(q_j).$$

¹⁴Our choice of beta-distributed priors follows the data patterns reported in Appendix Figure 5.

¹⁵This assumption is supported by descriptive evidence in Appendix Table 6 showing that higher-quality products are more likely to be regulated.

Suppose that woman i can be either informed ($v = I$) or uninformed ($v = U$) about product quality. Conditional on the signal and the set of available products, women form Bayesian posterior beliefs about the quality of contraceptive j . For informed women, $v = I$, posterior beliefs coincide with true product quality, $E_{It}[q_j|J_{ht}, r_{jt}] = \theta_j$. For uninformed women, $v = U$, given the distributional assumptions, the mean of the posterior distribution is

$$E_{Ut}[q_j|J_{ht}, r_{jt}] = \sigma E[q_j|J_{ht}] + (1 - \sigma)r_{jt},$$

where $\sigma = (\mu + \omega)/(\mu + \omega + 1)$.

We model contraceptive choice among women who wish to avoid pregnancy using a random coefficients logit model. The indirect utility of woman i from choosing contraceptive j in month t is

$$u_{ijt} = (\alpha_i + \tau \cdot \mathbb{1}\{j \in \text{INS}\}) \log(p_{jt}^{\text{eff}}) + \gamma E_{v(i)t}[q_j|J_{ht}, r_{jt}] + \rho_i \mathbb{1}\{j \in \text{IUD}\} + \xi_{jt} + \varepsilon_{ijt},$$

where

$$\alpha_i = \alpha_0 + \mathbf{D}_i' \boldsymbol{\alpha}_d$$

$$\rho_i = \rho_0 + \rho_1 \cdot \text{age}_i$$

$\mathbf{D}_i = (\text{age}_i, \text{low income}_i, \text{INS history}_i)'$, “low income” takes the value of one if the woman makes less than a 2018 monthly minimum wage and “INS history” is the probability that the woman obtains contraceptives in the insurance segment, which we predict based on her age from the DHS. The variable $p_{jt}^{\text{eff}} = \{p_{jt}, w_{jt}\}$ is the effective price the woman faces for product j , which is equal to the retail price p_{jt} when the product is supplied OTC and equal to the wholesale price w_{jt} when the product is supplied through insurance.¹⁶ $E_{v(i)t}[q_j|J_{ht}, r_{jt}]$ is the woman’s posterior belief about product j ’s quality, and $\mathbb{1}\{j \in \text{IUD}\}$ is an indicator for IUDs.

The term ξ_{jt} is the unobserved product characteristic. We decompose this term as

¹⁶Although IUDs in Colombia can be purchased over the counter, women still need to see a healthcare provider to have the device inserted. Hence, we allow both the effective price and the hassle cost to affect the choice of IUDs offered OTC.

$\xi_{jt} = \xi_j + \xi_t + \Delta\xi_{jt}$, where the first component captures preferences for time-invariant contraceptive j 's characteristics, the second captures time-shocks, and the third is a structural demand error. We normalize the utility of not buying contraceptives to zero, so that $u_{i0t} = \varepsilon_{ijt}$, and assume that ε follows a type-I extreme value distribution.

Define the mean utility and the woman-specific component of utility, respectively, as:

$$\begin{aligned}\delta_{jt} &= (\alpha_0 + \tau \cdot \mathbb{1}\{j \in \text{INS}\}) \log(p_{jt}^{\text{eff}}) + \rho_0 \mathbb{1}\{j \in \text{IUD}\} + \xi_j + \xi_t + \Delta\xi_{jt}, \\ \mu_{ijt} &= (\mathbf{D}_i' \boldsymbol{\alpha}_d) \cdot \log(p_{jt}^{\text{eff}}) + \gamma E_{v(i)t}[q_j | J_{ht}, r_{jt}] + \rho_1 \mathbb{1}\{j \in \text{IUD}\} \cdot \text{age}_i.\end{aligned}\quad (1)$$

Then, the model's predicted market share of contraceptive j is:

$$\hat{s}_{jt} = \int s_{ijt} dP(v, a), \quad (2)$$

where s_{ijt} is the probability that woman i chooses contraceptive j , which is given by:

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{|J_t|} \exp(\delta_{kt} + \mu_{ikt})}.$$

6.1.1 Identification

We estimate two types of parameters, one associated with the mean utility and the other with the woman-level component of utility: $\boldsymbol{\Phi} = (\boldsymbol{\Phi}^l, \boldsymbol{\Phi}^{nl})' = ((\alpha_0, \tau, \rho_0), (\boldsymbol{\alpha}_d', \rho_1, \gamma, \sigma))'$.

A. Linear parameters, $\boldsymbol{\Phi}^l$. To identify the linear parameters, we first fix $\boldsymbol{\Phi}^{nl}$ and recover the mean utility term δ_{jt} by iterating on a contraction mapping between model-predicted market shares in equation (2) and observed market shares (Berry et al., 1995). Then, we estimate equation (1) and address price endogeneity using as instrument for the effective price of product j and its interaction with the insurance segment indicator the predicted percentage change in prices for targeted products after the regulation (as in Figure 2, Panel A) and an indicator for targeted products after the regulation.

B. Non-linear parameters, Φ^{nl} . We discuss identification of the nonlinear parameters beginning with beliefs and preferences for quality and then turning to demographic interactions.

B.1. Beliefs and preference for quality. A key identification challenge in our model with information frictions is to separately identify preferences for quality, captured by γ , from imperfect information about quality, captured by σ , since their effects on women's choices are observationally equivalent. To separately identify preferences from information frictions, we exploit the individual-level insurance data to construct two micro-moments that target the difference-in-differences estimator from Figure 4.¹⁷ Intuitively, the fraction of informed women (those with higher healthcare spending) who consume products targeted by the price ceiling policy identifies the preference for quality, while the difference in the fraction of uninformed women (those with lower healthcare spending) who consume targeted products before and after the policy identifies the relative weight women place on information signals. The respective micro-moments are:

$$\begin{aligned} \text{For } \gamma: \quad g^{\text{qual}}(\Phi) &= \frac{1}{T} \sum_{t=1}^T \widehat{m}_t^{\text{qual}}(\Phi) - \overline{m}^{\text{qual}} = 0, \\ \text{For } \sigma: \quad g^{\text{reg}}(\Phi) &= \left(\frac{1}{T_{\text{post}}} \sum_{t \in \text{post}} \widehat{m}_t^{\text{reg}}(\Phi) - \frac{1}{T_{\text{pre}}} \sum_{t \in \text{pre}} \widehat{m}_t^{\text{reg}}(\Phi) \right) - (\overline{m}_{\text{post}}^{\text{reg}} - \overline{m}_{\text{pre}}^{\text{reg}}) = 0, \end{aligned}$$

with:

$$\widehat{m}_t^{\text{qual}}(\Phi) = \frac{\frac{1}{R} \sum_{i,j} \mathbb{1}\{j \in \text{reg}\} \cdot \mathbb{1}\{v_i = I\} \cdot \hat{s}_{ijt}(\Phi)}{\frac{1}{R} \sum_{i,j} \mathbb{1}\{v_i = I\} \cdot \hat{s}_{ijt}(\Phi)},$$

¹⁷This indirect inference approach follows [Fu and Gregory \(2019\)](#) who match regression discontinuity estimates.

$$\widehat{m}_t^{\text{reg}}(\Phi) = \frac{\frac{1}{R} \sum_{i,j} \mathbb{1}\{j \in \text{reg}\} \cdot \mathbb{1}\{v_i = U\} \cdot \hat{s}_{ijt}(\Phi)}{\frac{1}{R} \sum_{i,j} \mathbb{1}\{v_i = U\} \cdot \hat{s}_{ijt}(\Phi)},$$

where $\mathbb{1}\{j \in \text{reg}\}$ is an indicator for whether product j is targeted by the price ceiling policy; R denotes the number of women sampled for estimation, set to 1000; and T denotes the number of months. The empirical targets $\overline{m}^{\text{qual}}$, $\overline{m}_{\text{post}}^{\text{reg}}$ and $\overline{m}_{\text{pre}}^{\text{reg}}$ are computed from the claims data.

B.2. Demographic interactions. To identify the parameters capturing individual heterogeneity in preferences, we use another set of micro-moments capturing the consumption patterns by age and route of administration:

$$g^{A,C}(\Phi) = \frac{1}{T} \sum_{t=1}^T \widehat{m}_t^{A,C}(\Phi) - \overline{m}^{A,C} = 0, \quad g^{D,w}(\Phi) = \frac{1}{T} \sum_{t=1}^T \widehat{m}_t^{D,w}(\Phi) - \overline{m}^{D,w} = 0,$$

with

$$\widehat{m}_t^{A,C}(\Phi) = \frac{\frac{1}{R} \sum_{i,j} \text{age}_i \cdot C_j \cdot \hat{s}_{ijt}(\Phi)}{\frac{1}{R} \sum_{ij} C_j \cdot \hat{s}_{ijt}(\Phi)}, \quad \widehat{m}_t^{D,w}(\Phi) = \frac{\frac{1}{R} \sum_{i,j \in N_{\text{ins}}} \log(w_{jt}) \cdot D_i \cdot \hat{s}_{ijt}(\Phi)}{\frac{1}{R} \sum_{i,j \in N_{\text{ins}}} D_i \cdot \hat{s}_{ijt}(\Phi)}$$

where $C_j \in \{\text{pills, injections, IUDs}\}$, $\mathbf{D}_i = (D_{i1}, D_{i2}, D_{i3})'$ with $D_{i1} = \mathbb{1}\{\text{age}_i > 35\}$, $D_{i2} = \text{low income}_i$, $D_{i3} = \mathbb{1}\{\text{INS history}_i \geq 0.44\}$, N_{ins} is the number of product-month observations in the insurance segment, and the targets $\overline{m}^{A,C}$ and $\overline{m}^{D,w}$ are computed from the health claims data.

Finally, we bring the matrix of exogenous variables from the inner loop that identifies

Φ^l , namely $\mathbf{Z}_d$, to the outer loop to form the moment condition:

$$\mathbf{g}^{\text{agg}}(\Phi) = \frac{1}{N} \mathbf{Z}_d' \xi(\Phi) = \mathbf{0}$$

where N is the total number of product-month observations. These moments help identify Φ^{nl} through δ_{jt} .

6.2 Estimation Approach

We use a generalized method of moments (GMM) estimator that combines standard demand-side instruments with micro-moments derived from the price ceiling policy and our individual-level health claims data. Collecting all the moment conditions presented above in the vector $\mathbf{g}(\Phi)$, our GMM estimator solves the following objective function:

$$\hat{\Phi} = \arg \min_{\Phi} \mathbf{g}(\Phi)' \mathbf{W}_d^{-1} \mathbf{g}(\Phi)$$

The weighting matrix $\mathbf{W}_d$ is block-diagonal, with each block equal to the empirical variance of the corresponding micro-moment and a block equal to $\mathbf{Z}_d' \mathbf{Z}_d$ for the aggregate moment conditions. Because the two types of moments are computed from samples of different sizes, we rescale the aggregate moment conditions to prevent one block from dominating the objective function (Conlon and Gortmaker, 2020).

6.3 Supply

The OTC contraceptive supply chain involves price negotiations between manufacturers and wholesalers and between wholesalers and retail pharmacies (see Figure 1). However, we do not observe these negotiated prices in the data. Because IQVIA reports only retail prices, we model manufacturers in the OTC segment as setting retail prices as if they were vertically integrated with pharmacies. This assumption is reasonable since retail pharmacy markets in Colombia are highly competitive.

In the insurance segment, manufacturers and/or wholesalers sell to health insurers and negotiate prices. Because the OTC segment is manufacturers' largest and most profitable

segment, we assume that manufacturers are willing to supply insurers only if they can capture most of the gains from trade. We assume, for simplicity, that manufacturers contract directly with insurers and possess all the bargaining power. We further assume that there is a single representative insurer. Manufacturers make take-it-or-leave-it (TIOLI) offers to the insurer, specifying a per-unit wholesale price. The insurer can accept or reject the offer depending on whether the offer satisfies its participation constraint. Rejecting an offer yields zero surplus to the insurer.¹⁸

Manufacturer f chooses retail prices in the OTC segment and wholesale prices in the insurance segment to maximize total profits given by:

$$\Pi_{ft}(\mathbf{p}^{\text{eff}}) = \sum_{j \in J_{ft, \text{otc}}} (p_{jt} - mc_{jt}) \cdot s_{jt}(\mathbf{p}^{\text{eff}}) M_t + \sum_{k \in J_{ft, \text{ins}}} (w_{kt} - mc_{kt}) \cdot s_{kt}(\mathbf{p}^{\text{eff}}) M_t$$

where $J_{ft,h}$ is the set of products manufactured by firm f in month t and segment h , $\mathbf{p}^{\text{eff}}$ is the vector of effective prices across all firms, mc_{jt} is the marginal cost of product j in month t , s_{jt} is the product's market share, and M_t is the potential market size. We calibrate M_t using contraceptive prevalence estimates from the DHS, which the World Bank uses to estimate that 83% of reproductive-age women in Colombia use contraception ([World Bank, 2025](#)). Hence, excluding women who are sterilized, the outside option of not using hormonal contraceptives has a market share of roughly 40%.¹⁹

In the OTC segment, contraceptive manufacturers compete on retail prices in a Nash–Bertrand fashion. In the insurance segment, manufacturers choose wholesale prices to maximize profits subject to the insurer's participation constraint. Following [Gowrisankaran et al. \(2015\)](#), we write the insurer's surplus function as

$$U_t(J_{t, \text{ins}}; \varphi) = -E[\alpha_i]^{-1} M_t \underbrace{\int \mathcal{I}_{it, \text{ins}} dP_{l(i)}(v, a)}_{\tilde{\mathcal{I}}_t} - \varphi_t M_t \sum_{k \in J_{t, \text{ins}}} w_{kt} s_{kt}(\mathbf{p}^{\text{eff}}),$$

¹⁸There are no incentive compatibility constraints for the insurer because the set of products covered by the national health insurance plan is exogenously determined by the government.

¹⁹In our estimation, the market share of the outside option varies between 35% and 45% across months.

where $\mathcal{I}_{it,ins}$ is the individual-specific inclusive value from the logit demand, given by

$$\mathcal{I}_{it,ins} = \log \left(\sum_{j \in J_{t,ins}} \exp(\delta_{jt} + \mu_{ijt}) \right).$$

The term $\bar{\mathcal{I}}_t$ denotes the inclusive value aggregated across consumers. We convert it into monetary units by dividing by the absolute value of the average price coefficient in the insurance segment, $-E[\alpha_i]^{-1}$, and scale it by market size, M_t . The parameter $\varphi_t > 0$ captures the weight the insurer places on total contraceptive expenditures relative to women's expected utility. The insurer offers the bundle $J_{t,ins}$ to its enrollees whenever its surplus is nonnegative. We treat this product bundle as exogenous because it is determined by the government.

The Lagrangian of the manufacturer's problem is as follows:

$$\max_{p_{jt}, w_{kt}} \mathcal{L} = \sum_{j \in J_{ft,otc}} (p_{jt} - mc_{jt}) s_{jt}(\mathbf{p}^{\text{eff}}) M_t + \sum_{k \in J_{ft,ins}} (w_{kt} - mc_{kt}) s_{kt}(\mathbf{p}^{\text{eff}}) M_t + \lambda (U(J_{ins,t}; \varphi) - 0) \quad (3)$$

where λ captures the marginal value of insurer's profits to the manufacturer. Our equilibrium concept is a Nash equilibrium. Profit maximization involves a set of first-order conditions (FOCs) that after stacking and writing in matrix notation yields:

$$\mathbf{0} = \begin{bmatrix} \mathbf{s}_{otc} \\ \mathbf{s}_{ins} \end{bmatrix} + \begin{bmatrix} \mathbf{\Lambda}_{p,otc} & \mathbf{\Lambda}_{p,ins} \\ \mathbf{\Lambda}_{w,otc} & \mathbf{\Lambda}_{w,ins} \end{bmatrix} \begin{bmatrix} \mathbf{p} - \mathbf{mc}_{otc} \\ \mathbf{w} - \mathbf{mc}_{ins} \end{bmatrix} + \lambda \left(\begin{bmatrix} \mathbf{0} \\ \nabla_w \bar{\mathcal{I}} \end{bmatrix} - \varphi \begin{bmatrix} \nabla_p C \\ \nabla_w C \end{bmatrix} \right)$$

Here, $\mathbf{\Lambda}$ is the matrix of demand derivatives with respect to prices multiplied by the ownership matrix, $\nabla_w \bar{\mathcal{I}}$ is the derivative of the inclusive value with respect to the wholesale price, and $\nabla_w C$ and $\nabla_p C$ are the derivatives of total contraceptive costs in the insurance segment with respect to the wholesale price and the retail price, respectively. The manufacturer's optimization problem in equation (3) applies to the months before the price ceiling policy. After the policy, the derivatives of OTC contraceptive demand for targeted products with respect to prices are zero, since the price ceiling is binding.

The FOC of the manufacturer's problem yields an expression for the implied marginal

cost of each product:

$$mc_{jt} = p_{jt}^{\text{eff}} + \Delta_{jt}^{-1} s_{jt} + \lambda \Delta_{jt}^{-1} \nabla U_t$$

We decompose the marginal cost as $mc_{jt} = x'_{jt} \beta^s + \omega_{jt}$, where x_{jt} are observed product characteristics including the product's number of cycles, true quality, age, and whether it is imported; β^s are supply-side parameters to be estimated; and ω_{jt} is the supply-side structural error capturing unobserved cost shifters. Substituting, we obtain the estimating equation:

$$x'_{jt} \beta^s + \omega_{jt} = p_{jt}^{\text{eff}} + \Delta_{jt}^{-1} s_{jt} + \lambda \Delta_{jt}^{-1} \nabla U_t \quad (4)$$

Additional considerations. We do not model how segmentation arises from the OTC approval process or from the determination of insurance formularies. Segmentation could reflect the selection of manufacturers or products into each segment, or differences in the women each segment serves. We take the first two as given and focus on how women's demand responds to the resulting market structure. Two features of the setting support this. First, we do not model manufacturers' participation in each segment because it does not change over our sample period. Second, the composition of products within each segment is likewise stable because there are no new OTC approvals or changes to the insurance formulary over the sample period. Still, one of our counterfactuals allows manufacturers to discontinue products that become redundant once the market is pooled, which speaks to the role of endogenous product composition.

Identification and Estimation Approach. We estimate the supply parameters $\Theta = (\lambda, \beta^s, \varphi_t)'$ using GMM, taking demand estimates as given (as in [Miller and Weinberg, 2017](#)) and restricting estimation to the pre-policy period from January to December 2018.

The parameter φ_t is identified directly from the insurer's participation constraint. In equilibrium this participation constraint must be binding, otherwise the manufacturer can offer a slightly higher price to the insurer, raising profits and still satisfying the participation constraint. Setting the participation constraint to zero therefore yields a

closed-form expression for φ_t :

$$\varphi_t = \frac{-E[\alpha_i]^{-1} \bar{I}_t}{\sum_{k \in J_{t,ins}} w_{kt} s_{kt}(\mathbf{p}^{\text{eff}})}$$

The linear parameters β^s are identified from equation (4) under the assumption that the product characteristics in x_{jt} are exogenous to unobserved cost shifters.

Finally, identification of λ requires an instrumental variable approach. This Lagrange multiplier is potentially endogenous because it reflects the manufacturer's optimal response to cost conditions that are observed by the manufacturer but unobserved to the econometrician. To address this, we instrument the insurer's surplus with two sets of instruments. The first are [Gandhi and Houde \(2020\)](#)-style differentiation instruments, constructed from the grams of estrogen and progestin in rival products per month. The second is the aggregate demand of rival manufacturers in the OTC segment. Both instruments satisfy the exclusion restriction for distinct reasons. The hormonal composition of a contraceptive is set by pharmacological requirements and is therefore orthogonal to a manufacturer's idiosyncratic cost shocks. Rival manufacturers' OTC demand shifts the focal manufacturer's outside option in its TIOLI offer to the insurer without directly affecting the focal product's own marginal cost.

We implement these instruments in our GMM procedure by constructing the moment condition:

$$\mathbf{g}^{\text{supply}}(\Theta) = \frac{1}{N_{\text{pre}}} \mathbf{Z}'_s \omega(\Theta) = \mathbf{0}$$

where the matrix $\mathbf{Z}_s$ includes the exogenous product characteristics x_{jt} and our instruments, and N_{pre} is the number of product-month observations in the pre-policy period. The GMM estimator solves the following objective function:

$$\hat{\Theta} = \arg \min_{\Theta} \mathbf{g}(\Theta)' \mathbf{W}_s^{-1} \mathbf{g}(\Theta)$$

where $\mathbf{W}_s = \mathbf{Z}'_s \mathbf{Z}_s$. Finally, to ensure that the Kuhn-Tucker condition $\lambda \geq 0$ is satisfied, we reparametrize $\lambda = \exp(\tilde{\lambda})$ and estimate $\tilde{\lambda}$ directly. Since $\tilde{\lambda}$ in principle can take very negative values, a significance test on λ will inform whether the model rejects pure

Bertrand pricing in the insurance segment. Appendix 4 provides the estimation details.

7 Results

Table 2 reports our estimation results. Women experience negative marginal utility from both retail prices and wholesale prices, but the disutility from wholesale prices in the insurance segment is substantially larger, suggesting prices and hassle costs deter contraceptive take-up. Price sensitivity is higher among women who are younger, lower income, and more likely to have claimed contraceptives through insurance. The median own-price elasticity is -1.73 in the OTC segment and -3.71 in the insurance segment; this elasticity difference is what allows the manufacturer to price-discriminate across segments.

Women have strong preferences for higher-quality contraceptives, but they face information frictions about quality: they place 20% weight on their prior beliefs about product quality and 80% on new quality signals, indicating that most women remain poorly informed about contraceptive quality in the absence of a policy shock. Finally, the relative appeal of IUDs increases with age.

The second panel of Table 2 reports the supply-side estimates. Marginal costs are lower for imported products than for domestically produced ones, in line with multinational companies being more efficient than local producers. Products covering more menstrual cycles have lower marginal costs per cycle, while products introduced earlier in the market have higher marginal costs than their counterparts. These parameters are imprecise but informative about the cost structure of contraceptive manufacturers. Imposing the insurer's participation constraint to recover φ implies that insurers place a weight of 60% on total contraceptive costs and the remaining 40% on women's surplus.²⁰ The estimated Lagrange multiplier λ is statistically significant, allowing us to reject pure Bertrand pricing in the insurance segment. Appendix Table 7 shows that our model closely matches the empirical micro-moments from the claims data.

²⁰We recover the explicit weight on contraceptive costs by solving for ψ from $\varphi = \frac{\psi}{1-\psi}$.

TABLE 2: Model Estimation Results

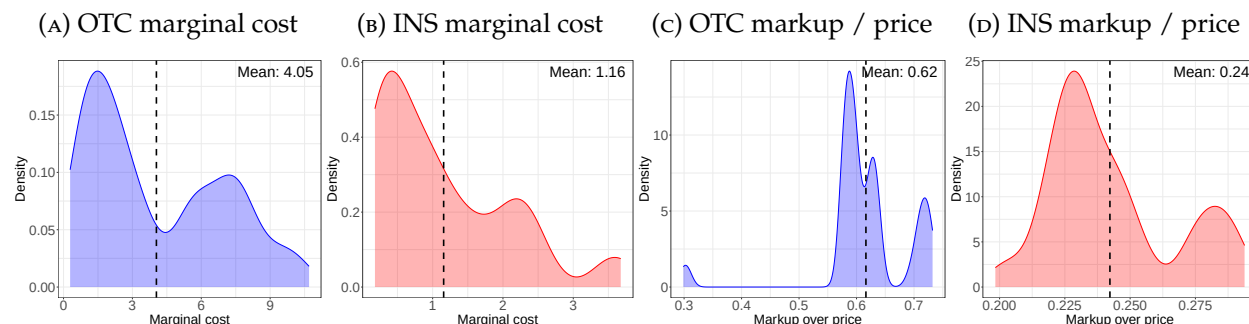
Variable	coef	se (·) sd [·]
<i>Panel A. Demand parameters</i>		
Log price	-1.643***	(0.243)
Log price x Age	0.072**	(0.035)
Log price x Low income	-0.012**	(0.005)
Log price x INS history	-0.223***	(0.029)
Log price x INS dummy (Hassle cost)	-1.987***	(0.008)
IUD dummy x Age	15.74***	(0.350)
Quality preference (γ)	0.506*	(0.298)
Weight on prior (σ)	0.196	(0.492)
<i>Panel B. Supply parameters (pre-period)</i>		
Number of cycles	-0.006	(0.070)
Imported product	-0.194	(0.444)
Product age	0.019	(0.220)
Shadow price of insurance (λ)	0.011*	(0.007)
Weight on insurer costs (φ)	0.592	[0.021]
<i>Panel C. Model metrics</i>		
Median own-price elasticity (OTC)	-1.727	[0.363]
Median own-price elasticity (INS)	-3.710	[0.033]
Demand GMM Objective	0.016	
Supply GMM Objective	44.694	
<i>Demand Fixed Effects</i>		
ATC, Route, Nest, Quarter	Yes	
Observations (product-months)	1728	
<i>Supply Fixed Effects</i>		
ATC, Route, Nest, Lab, Quarter	Yes	
Observations (product-months)	864	

Note: Table presents the demand and supply model estimates. In the demand model we use as instruments for the log effective price and its interaction with the insurance segment dummy the predicted percentage change in prices for targeted products after the regulation and an indicator for targeted products after the regulation. We recover the weight insurer's place on contraceptive costs by setting the participation constraint to zero. The instruments for the shadow price of insurance are the average distance between the focal product and its competitor's grams of estrogen and progestin and the demand for other manufacturers in the OTC segment excluding the focal manufacturer. Age is divided by 100 in estimation. The regression for the mean utility includes ATC, route, segment, and quarter fixed effects. The regression for the marginal cost includes ATC, route, segment, manufacturer, and quarter fixed effects. The demand model is estimated on data from January 2018 to December 2019. The supply model is estimated on data from January to December 2018. Standard errors are reported in parenthesis and standard deviations in brackets.

Model metrics. Appendix Table 8 reports average own- and cross-price elasticities in the OTC segment during the pre-policy period by route of administration. Own-price elasticities exceed one in absolute value across all routes and fall within the ballpark of those derived from experimental variation in out-of-pocket contraceptive prices in the U.S. (Bailey et al., 2025). Demand for conventional pills is more elastic than for injections, and

IUD users are more likely to switch to injections than to conventional pills when prices rise. Appendix Table 9 reports hassle cost elasticities for the insurance segment, which are much larger in magnitude than price elasticities in the OTC segment. Substitution patterns in the insurance segment show that IUD users are more likely to switch to implants than injections.

FIGURE 6: Distribution of Marginal Costs and Markups



Note: Figure presents the distribution of marginal costs and markups separately by market segment. The supply model estimation uses data from January to December 2018. The dashed line represents the mean of the distribution.

Figure 6 presents the implied distributions of marginal costs and markups across both segments. Marginal costs are largely bimodal in both segments because of the costs of the two most common active ingredients, levonorgestrel and medroxyprogesterone. All estimated marginal costs are positive, suggesting our model fits the data well. The (unweighted) average marginal cost is \$4.05 in the OTC segment, reflecting the higher quality of some products offered exclusively there, and \$1.16 in the insurance segment.

Retail markups also have a bimodal distribution with the average equal to 62%. In the insurance segment, estimated wholesale markups are on average 24%. These two estimates imply that if the manufacturer sold insurance-exclusive products OTC, it would earn a markup of nearly 4 times the marginal cost of production.²¹

²¹This estimate is computed as the difference between the average OTC retail price and the average marginal cost in the insurance segment, divided by the average marginal cost in the insurance segment.

8 Equilibrium Impacts of Market Segmentation

Counterfactual setup. We now use our model estimates to evaluate the equilibrium effects of market segmentation. Later in this section, we extend our analysis to include impacts on fertility. We consider two main counterfactuals, an OTC-only market and an insurance-only market, and report results holding information status fixed (labelled DS) and assuming full information (labelled DSI).

In the OTC-only DS counterfactual, we set the insurance segment indicator to zero in the demand function, hence hassle costs have no impact on demand except for IUDs, which puts upward pressure on equilibrium prices. On the supply side, manufacturers price all products according to Nash-Bertrand competition. Two opposing forces shape equilibrium prices in this counterfactual. Removing the insurer's participation constraint allows manufacturers to solve an unconstrained pricing problem, which tends to raise prices. At the same time, manufacturers must offer all insurance-segment products through the OTC segment, intensifying competition across products and putting a downward pressure on prices. The net effect is therefore ambiguous a priori.

In the DSI counterfactual, we additionally remove information frictions by assuming all women are fully informed. The equilibrium price effects of better information depend on the direction of the quality bias. If women systematically underestimate the quality of higher-priced products, then removing information frictions shifts demand toward these products, putting upward pressure on their prices. Conversely, if information frictions disproportionately affect lower-quality products, removing frictions would shift demand away from these products, putting downward pressure on their prices. The net effect therefore depends on the distribution of quality misperceptions across products, which we recover from the structural estimates.

The insurance-only counterfactual follows an analogous procedure. On the demand side, we set the insurance segment indicator to one, so hassle costs impact demand for all products, putting downward pressure on equilibrium prices. On the supply side, manufacturers price all products to satisfy the insurer's participation constraint, which tends to lower prices. However, because OTC products have higher marginal costs, pooling

the market may raise prices relative to baseline.

Equilibrium. Our estimates describe the observed equilibrium, which raises the question of how far the counterfactuals extrapolate from it. Although we do not prove existence or uniqueness of the counterfactual price equilibrium, we verify numerically that the system of first-order conditions converges to the same fixed point from a range of starting values, as is standard in the literature.

Results. We simulate all counterfactuals in the period between January and June 2018 before the announcement of the price ceiling policy. To ensure comparability across scenarios, we report counterfactual price results separately for products that are offered OTC at baseline in Table 3 and for products that are offered through insurance at baseline in Table 4. By reporting results in this way we avoid comparisons of retail and wholesale prices.

Table 3 shows that moving to an OTC-only market in the DS scenario slightly lowers the average price. The effect of a lower price elasticity of demand and of greater product competition therefore nearly offset each other in this counterfactual. We find similarly modest reductions in prices across pills and implants, but prices for IUDs and injections increase 7% and 2%, respectively. Incorporating full information in column (3) further reduces average equilibrium prices, suggesting that manufacturers exploit these frictions to sustain higher markups and that women's prior beliefs about quality tend to be biased upward.

Although the price reductions in the OTC-only counterfactuals are modest in percentage terms, they are economically meaningful at the market level. Holding demand fixed, the price decrease under the DSI scenario would reduce annual out-of-pocket spending on contraceptives by 5.2 billion pesos.²² Hence, market segmentation allows manufacturers to price-discriminate and charge higher retail prices for products offered OTC at baseline, in line with the reduced form evidence.

²²We calculate this figure by multiplying the price change in dollars by the COP/USD exchange rate and by total annual demand in the observed scenario.

TABLE 3: Counterfactual Retail Prices in Baseline OTC Segment

Route	Observed (1)	DS (2)	DSI (3)
All types	5.831	5.802	5.743
Conventional Pill	8.594	8.445	8.416
Emergency Pill	3.105	3.054	3.058
IUD	2.244	2.407	2.404
Implant	1.114	1.112	1.112
Injection	4.374	4.456	4.493

Note: Table presents the demand-weighted average retail price conditional on products offered OTC at baseline. We report several scenarios: the observed scenario (column 1); the OTC-only counterfactual holding information status fixed (column 2); and the OTC-only counterfactual assuming full information (column 3). All counterfactual simulations use information from January to June 2018.

TABLE 4: Counterfactual Wholesale Prices in Baseline INS Segment

Route	Observed (1)	DS (2)	DSI (3)
All types	0.821	0.844	0.844
Conventional Pill	1.054	1.050	1.051
IUD	1.394	1.444	1.443
Implant	0.673	0.699	0.699
Injection	0.803	0.826	0.826

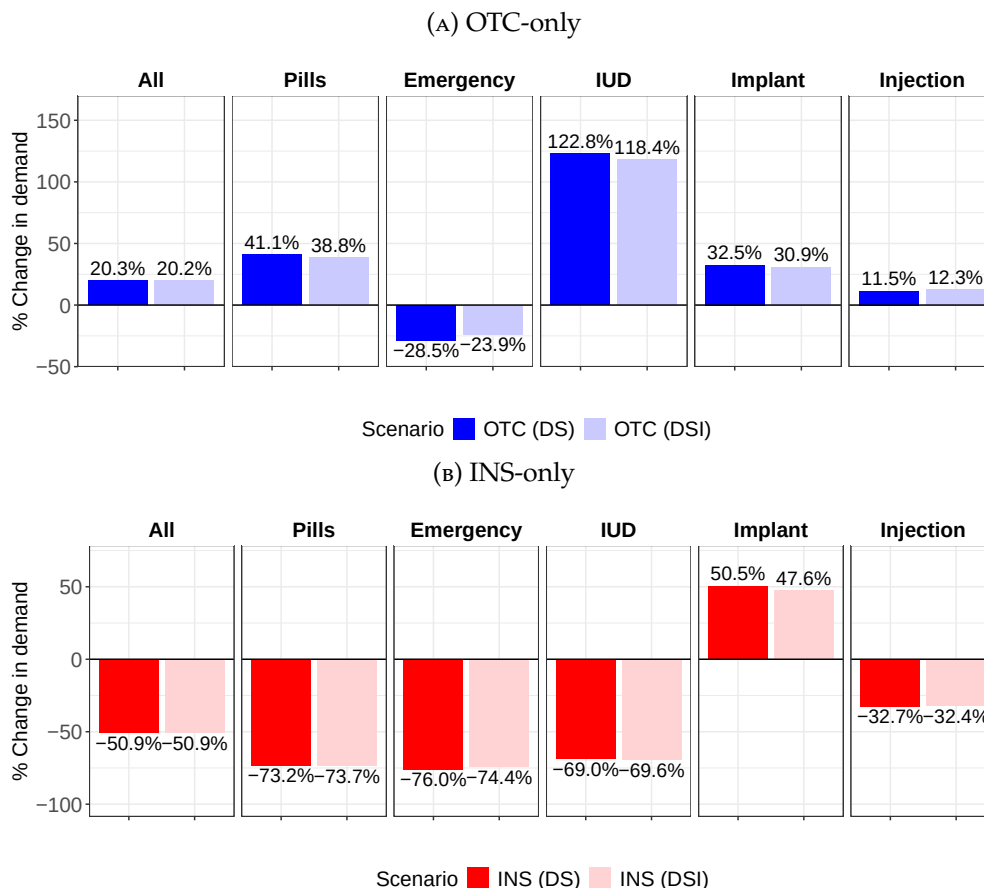
Note: Table presents the demand-weighted average wholesale price conditional on products offered INS at baseline. We report several scenarios: the observed scenario (column 1); the INS-only counterfactual holding information status fixed (column 2); and the INS-only counterfactual assuming full information (column 3). All counterfactual simulations use information from January to June 2018.

In Table 4, moving to an insurance-only market in the DS scenario raises overall wholesale prices by 3% relative to the observed scenario. Increases are similar across routes of administration, except for conventional pills, whose prices remain roughly constant. The price increase suggests that the insurer's surplus is not large enough to prevent manufacturers from passing through the higher marginal costs of products offered OTC at baseline. Removing information frictions in column (3) generates a similar price increase as column (2). Scaled to the market level, these wholesale price changes imply that insurers would pay 1.3 billion pesos more annually for contraceptives.

Appendix Figure 6 plots the counterfactual price trends over the pre-policy period. The patterns are stable across months, confirming that the average effects reported in Tables 3 and 4 are not driven by seasonal variation in contraceptive demand. Appendix Tables 10 and 11 report diversion ratios to the outside option in each counterfactual and

the observed scenario. Consistent with the previous results on prices, diversion ratios increase in the insurance-only market but decrease in the OTC-only market.

FIGURE 7: Counterfactual Demand Changes



Note: Figure presents the percentage change in demand for contraceptives relative to the observed scenario in the OTC-only counterfactual (Panel A) and the INS-only counterfactual (Panel B). Dark colors represent the scenario holding information status fixed. Light colors represent the scenario with full information. Counterfactual simulation use data from January to June 2018.

Figure 7 presents the percentage change in contraceptive demand relative to the observed scenario for the OTC-only counterfactuals (Panel A) and the INS-only counterfactuals (Panel B). Moving to an OTC-only market in the DS scenario raises total demand by 20%, driven by a 41% increase in demand for conventional pills. Demand for emergency pills falls 28%, suggesting these products partly serve women who would otherwise obtain contraceptives through insurance and who switch to conventional methods once those become available over the counter. Demand for IUDs rises 123%—albeit from a small baseline—even though their prices increase in this counterfactual, because women

substitute toward IUDs from injections, whose prices also rise. Assuming full information (DSI) yields slightly smaller increases than the DS scenario, but still sizable. In absolute terms, these increases correspond to 936,000 additional women using contraceptives each month.²³

The DS scenario in the insurance-only market reduces total demand by 51% relative to baseline. Demand falls by roughly 70% for conventional pills, emergency pills, and IUDs, and by 32% for injections. Demand for implants rises 50%, but not enough to offset the declines elsewhere. Assuming women are fully informed about product quality (DSI) also reduces aggregate demand, by a similar magnitude than under the DS scenario.

The starkly different demand responses in the OTC-only and insurance-only counterfactuals reveal the importance of hassle costs. In the insurance-only counterfactual, all products incur the hassle costs causing aggregate demand to fall sharply. The decline is concentrated precisely where we expect it to be: conventional pills, which women currently obtain OTC in part to avoid the insurance hassle, account for most of the reduction. Conversely, the OTC-only counterfactual eliminates these hassle costs for products currently available only through insurance, leading demand to rise despite the loss of insurance coverage.

In particular, the OTC-only results speak to how much women would be willing to pay to avoid the hassle costs of insurance. Let $\tilde{w}_j = \ln w_j$ and $\tilde{p}_j = \ln p_j$, the price-related components of indirect utility in the observed scenario and the scenario without hassle cost are:

$$V_{ij}^{\text{obs}} = (\alpha_i + \tau) \tilde{w}_j \quad \text{and} \quad V_{ij}^{\text{cf}} = \alpha_i \tilde{p}_j.$$

Define woman i 's *reservation retail price* for product j , denoted p_{ij}^* , as the retail price that would leave her indifferent between buying the product over the counter and obtaining it through insurance as in the observed market. Setting $\alpha_i \tilde{p}_{ij}^* = (\alpha_i + \tau) \tilde{w}_j$ and solving yields

$$\tilde{p}_{ij}^* = \left(1 + \frac{\tau}{\alpha_i}\right) \tilde{w}_j, \quad \text{so} \quad p_{ij}^* = \exp\left[\left(1 + \frac{\tau}{\alpha_i}\right) \tilde{w}_j\right].$$

²³We obtain this figure by multiplying the size of the potential market (the number of reproductive-age women excluding those who are sterilized) by the percentage change in demand.

The reservation price is the highest retail price woman i would accept before preferring to return to the insurance segment. The value of avoiding the hassle cost is:

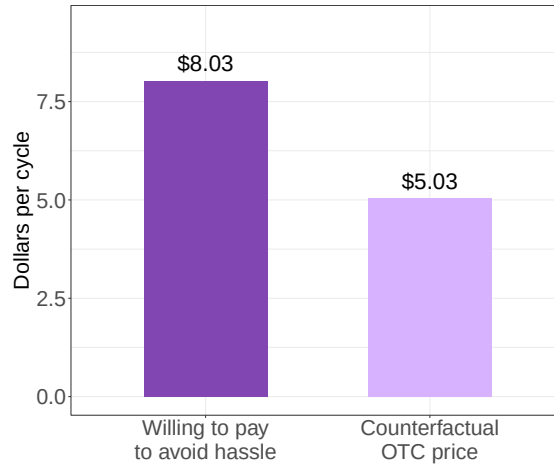
$$\text{WTP}_{ij}^{\text{hassle}} = p_{ij}^* - w_j, \quad (5)$$

and the additional amount the woman actually pays at retail in the OTC-only counterfactual is:

$$\Delta p_j = p_j^{\text{otc only}} - w_j. \quad (6)$$

Comparing (5) and (6) gives the net welfare consequence of the segment switch: when $\text{WTP}_{ij}^{\text{hassle}} > \Delta p_j$, the woman gains more from escaping the frictions of the insurance segment than she loses by buying over the counter, and the OTC-only counterfactual makes her better off.

FIGURE 8: Willingness to Pay to Avoid Hassle Costs



Note: Figure presents the reservation retail price that would make a women indifferent between buying OTC or through insurance in the observed scenario and the equilibrium price in the OTC-only counterfactual for products offered through insurance at baseline.

Figure 8 compares the average reservation retail price against the equilibrium retail price in the OTC-only counterfactual, for products purchased through insurance at baseline. Women who buy through insurance are willing to pay up to \$8 at retail to avoid the hassle of this segment, whereas the counterfactual OTC price of the products they choose at baseline is \$5, so they are better off switching to the OTC market. This willingness to pay is lower than the pre-regulation average price of a conventional pill in the OTC

segment (\$11), which explains why a fraction of sales still occurs through insurance in the observed market.

Our results have two implications. First, the hassle costs of obtaining contraceptives through insurance, rather than prices alone, are an important constraint on access. Second, they point to the value of policies that reduce these costs while preserving coverage, for example, allowing women to fill contraceptive claims at pharmacies rather than at insurer dispensing facilities. More broadly, the mechanism we identify may extend to other therapeutic classes segmented between insurance and OTC channels, where non-price frictions can similarly shape demand and access.

TABLE 5: Change in Consumer Surplus (Dollars per Woman-Year)

Route	OTC-only		INS-only	
	DS	DSI	DS	DSI
All	3.694	3.690	-8.664	-8.664
Low income	3.679	3.674	-8.617	-8.617
High income	3.739	3.734	-8.801	-8.801
Informed	3.661	3.667	-8.747	-8.748
Uninformed	3.717	3.705	-8.609	-8.607
Age 15-19	3.564	3.564	-8.488	-8.485
Age 20-35	3.618	3.615	-8.559	-8.558
Age 36+	4.068	4.051	-9.175	-9.180

Note: Table presents the change in women's surplus per capita measured in dollars per year between the OTC-only and INS-only counterfactuals and the observed scenario. Under each counterfactual, we report two scenarios: holding information status fixed (DS); and assuming full information (DSI). All counterfactual simulations use information from January to June 2018.

Additional outcomes. Table 5 reports the change in consumer surplus (in dollars per woman-year) between each counterfactual and the observed scenario. Consumer surplus is the inclusive value from the logit demand system, divided by the price coefficient α_i to convert to dollar terms. In the OTC-only counterfactual, the DS and DSI scenarios raise average consumer surplus by \$3.7, nearly 65% of the average counterfactual retail price. The gains are broadly shared across subgroups of women but somewhat larger for high-income and uninformed women. Consumer surplus also increases monotonically with age.

In the insurance-only market, consumer surplus falls by \$8.7. The welfare loss is more pronounced for informed women, consistent with their stronger preferences for higher-

quality products that now face hassle costs; and for high-income and older women who would rather pay the retail price. Imposing full information (DSI) reduces surplus by a similar amount.

9 Conclusion

This paper studies how market segmentation between over-the-counter (OTC) and insurance channels shapes access to pharmaceuticals with an application to hormonal contraceptives. We use data from Colombia where women can access contraceptives OTC at retail prices with zero hassle costs or through their health insurer at zero prices but with substantial hassle costs. First, we use a price ceiling policy introduced by the government in 2019 and a survey instrument to learn about the determinants of contraceptive demand. We show that contraceptive demand is significantly price-sensitive, that women face information frictions about product quality, and that women face hassle costs when sourcing contraceptives through insurance.

Next, to simulate the implications of removing market segmentation, we develop and estimate a structural model that takes into account the determinants of demand. Results show that market segmentation hurts contraceptive take-up relative to an OTC market and these effects are exacerbated when women lack accurate information about product quality. The reason is that market segmentation both allows pharmaceutical manufacturers to price discriminate and forces women to pay the cost of navigating the segments. Moving to an OTC-only market raises contraceptive demand by 20% and consumer surplus by \$3.7. Moving to an insurance-only market reduces demand by 50% and consumer surplus by \$8.7, as the hassle costs of interacting with the healthcare system outweigh the benefits of increased product competition. Results also imply that women would be willing to pay nearly \$8 to avoid the hassle of navigating the insurance segment. These findings have direct implications for OTC expansion in other drug classes and countries where pharmaceuticals are also offered through health insurance.

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Online Appendix

Manuscript Title: Segmentation in Pharmaceutical Markets: Evidence from Contraceptives

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Appendix 1 Details on Data Construction

1.1 IQVIA and INVIMA data

The IQVIA dataset reports monthly prices and quantities of contraceptives purchased OTC in Colombia between 2009 and 2023. We have information on all medications categorized under ATC codes G03A, G03C, and G03F.

An observation in the raw dataset is a combination of the national drug code (NDC), month, and pharmacy type. Pharmacy type indicates whether the report originates from a chain or stand-alone pharmacy. A pharmacy is classified as chained if it operates more than five establishments under the same corporate name. Independent pharmacies account for around 70% of OTC contraceptive sales in Colombia. The NDC uniquely identifies a drug and its commercial presentation. For example, a single oral contraceptive brand might have two presentations: one with a single blister containing 21 pills and another with two blisters or 42 pills.

Our data cleaning procedure proceeds as follows:

- *Sample selection from INVIMA.* We begin with the universe of products registered in INVIMA. We exclude NDCs corresponding to hormone replacement therapy (post-menopause) and treatments for amenorrhea. The resulting sample focuses on contraceptive methods, including conventional oral contraceptives, emergency pills, injections, implants, intrauterine devices (IUDs), and patches. The INVIMA dataset contains 1,285 NDCs corresponding to 324 products, where a product is defined as a commercial brand.
- *Imputation of NDCs in IQVIA data.* Due to inconsistencies in pack codes across datasets, we impute INVIMA NDC identifiers to IQVIA pack codes. This process

yields 227 matched NDCs out of 246 pack codes (corresponding to 190 products, including hormone replacement therapy). Unmatched observations account for less than 1.5% of total sales.

- *Merging INVIMA and IQVIA data.* We merge the INVIMA and IQVIA datasets using the matched identifiers, resulting in 183 NDCs corresponding to 162 products after excluding hormone replacement therapy.
- *Price normalization.* We deflate nominal prices using the Colombian Consumer Price Index (base: December 2018 = 100) and convert them to U.S. dollars using the 2018 average exchange rate (2,952 COP/USD).
- *Retail price adjustment.* As prices reported for independent pharmacies correspond to wholesale prices, we apply a 15% retail markup following IQVIA guidelines.
- *Standardization of units.* To ensure comparability across products with different packaging and dosage forms, we standardize prices and quantities to a single menstrual cycle.
- *Aggregation.* We aggregate the data to the product-month level.

After cleaning outliers, the final dataset comprises 156 unique products observed monthly from January 2009 to December 2023, yielding 16,551 observations in an unbalanced panel.

1.2 Enrollment and Health Claims Data

1.2.1 Product-level data

To construct the product-month panel for the insurance segment for the months between January 2018 and December 2019 we use individual-level health claims records. Below is the list of cleaning steps we use to construct this data:

Sample restrictions. We restrict the sample as follows:

- We retain only claims made by women of reproductive age (15–45).
- We restrict attention to outpatient prescription claims for hormonal contraceptives, identified using ATC codes and medication names from INVIMA.
- We drop claims with zero wholesale price.
- We determine the number of presentations per refill as the median number of months between refills per woman. For median refills over 3 months, we constrain to 3 months.
- We define a product as a brand-dosage combination. We standardize quantities and prices to a single menstrual cycle to ensure comparability across presentations. For products dispensed in multi-cycle presentations, we divide the price by the number of cycles and multiply the quantity accordingly.

Price cleaning. Wholesale prices in the health claims reflect negotiated contracts between insurers and providers and vary across insurers, providers, and contract types in ways unrelated to the underlying price of the contraceptive. Contract types include fee-for-service, capitation, and fee-for-package. We proceed in two steps to clean these prices:

1. We residualize observed log wholesale prices per claim on contract type–product, insurer–provider–product, and insurer–provider–month fixed effects to remove contract-driven price variation.
2. We define the “cleaned price” as the exponential of the residual of the previous regression, plus the constant, plus the fee-for-service dummy per product.
3. We winsorize the cleaned price at the 2nd and 98th percentiles to remove outliers, and define the final price as the median of the winsorized residual at the product-month level.

Market share construction. We aggregate cleaned claims to the product-month level, computing total cycles dispensed and the final price for each product-month cell.

1.2.2 Individual-level data

For descriptive analyses where we use the individual level data we follow the cleaning steps below.

Sample restrictions. We apply the following restrictions sequentially:

- We retain only women of reproductive age (15–45).
- To deal with the fact that there may be multiple prescriptions for contraceptives in the same day, we prioritize those with greater effectiveness or clinical recommendation following the table below, which was produced by ChatGPT.

APPENDIX TABLE 1: Suggested Ranking of Hormonal Contraceptives (High to Low Priority)

Rank	ATC Code	Reasoning
1	G03AD01	Implant – long-acting, high effectiveness
2	G03AC03	Depo-Provera – 3-month injectable
3	G03AC06	2-month injectable
4	G03AA14	Drospirenone progestin-only pill – newer, more selective
5	G03AA07	Desogestrel progestin-only pill – modern, widely used mini-pill
6	G03AA17	Nomegestrol progestin-only pill – selective, less common
7	G03AA12	Lynestrenol progestin-only pill – older generation
8	G03FA11	Combined oral contraceptive with Desogestrel – third-generation progestin
9	G03FA01	Combined oral contraceptive with Levonorgestrel – second-generation progestin

- We expand the annual enrollment data to a monthly panel, generating one observation per woman per month. We then merge in contraceptive claims, other health claims (outpatient, inpatient, and prescription), and income data from 2017 payroll records.
- Because contraceptives differ in their duration of protection, we carry forward contraceptive claims to fill months without a new prescription:
 - For three-month injections, we carry forward the claim for up to two additional months.
 - For monthly products (pills and one-month injections), we carry forward the claim for one additional month.

1.3 Construction of the Contraceptive Quality Measure

We construct a measure of contraceptive quality at the level of active ingredient, route of administration, and quarter using adverse event reports from the FDA Adverse Event Reporting System (FAERS) from 2014 to 2019. We describe the construction procedure below.

Data assembly. We download quarterly FAERS files covering patient demographics, drug, outcome, reaction, therapy, indication, and report source tables for each quarter from 2014Q1 to 2019Q4. We retain only cases in which the reported indication includes contraception-related terms (e.g., contraception, birth control, oral contraceptive, emergency contraception). For each case, we retain drugs listed as the primary or secondary suspect ($\text{role_cod} \in \{PS, SS\}$) and merge in the associated reaction and outcome records.

Severity classification. We classify adverse events into two categories:

- *High-severity events:* outcomes recorded in the FAERS outcomes table, including death, hospitalization, disability, and life-threatening events.
- *Low-severity events:* reactions recorded in the FAERS reactions table, including side effects such as weight gain, acne, mood changes, and irregular bleeding.

We collapse these counts to the active ingredient-route-year-quarter level, after standardizing drug names to active ingredients using a hand-coded crosswalk that maps brand names and generic names to their active ingredients and routes of administration.

Quality measure. We construct a quality measure based on the Proportional Reporting Ratio (PRR), a standard pharmacovigilance statistic that compares the reporting rate of adverse events for a given drug to the reporting rate across all other drugs in the same period. For active ingredient k in quarter q , the PRR is defined as:

$$\text{PRR}_{kq} = \frac{(a_{kq} + 0.5)/(a_{kq} + b_{kq} + 0.5)}{(c_{kq} + 0.5)/(c_{kq} + d_{kq} + 0.5)}$$

where a_{kq} is the number of high-severity adverse event reports for active ingredient k in quarter q , b_{kq} is the number of low-severity reports for the same ingredient, $c_{kq} = A_q - a_{kq}$ is the number of high-severity reports for all other active ingredients in the same quarter, and $d_{kq} = B_q - b_{kq}$ is the analogous count for low-severity reports. The 0.5 correction is applied to avoid division by zero for active ingredients with sparse reports. We then define true quality as:

$$\text{Quality}_{kq} = \frac{1}{1 + \text{PRR}_{kq}}$$

so that higher values correspond to lower relative adverse event rates and therefore higher quality. We collapse to the active ingredient-route-quarter level by taking the mean quality across cases, and carry forward values for quarters with missing observations using the most recent available value.

Imputation hierarchy. We construct the quality measure at four levels of aggregation, used hierarchically to impute missing values:

- *Level 1:* active ingredient combination and route of administration (e.g., Levonorgestrel + Ethinylestradiol, oral).
- *Level 2:* progestin component only and route of administration (e.g., Levonorgestrel, oral).
- *Level 3:* active ingredient combination only, pooled across routes.
- *Level 4:* progestin component only, pooled across routes.

For active ingredients approved in Colombia but not reported in FAERS, we supplement with adverse event reports from INVIMA, Colombia's national drug safety agency, using the same PRR methodology (*Level 5*).

Appendix 2 Survey

2.1 Survey Design

The following items are selected and translated from the Spanish-language survey instrument. Several of these questions are shown conditionally, based on responses to earlier questions in the survey; conditional logic is noted in italics where applicable.

Q4. How old are you? (*Shown only if respondent's sex assigned at birth is female.*) (Choose one answer.)

- Under 15
- 15–17
- 18–20
- 21–25
- 26–30
- 31–35
- 36–40
- 41–49
- 50 or more

Q7. Do you have children? (*Choose one answer.*)

- Yes
- No

Q8. Do you currently have a partner? (*Choose one answer.*)

- Yes, married
- Yes, in a common-law union (living together)
- Yes, has a partner but does not live together
- Does not currently have a partner

Q9. What is your current educational level? (*Choose one answer.*)

- Primary school
- Secondary school
- High school diploma
- Technical/vocational
- University (undergraduate)

- Graduate/postgraduate

Note: Indicate the highest level of education completed.

Q12. What is the socioeconomic stratum (*estrato*) on the electricity bill for the place where you currently live? (If your residence has no assigned stratum or is in a heritage/protected zone, indicate the one that most closely resembles it.) (*Choose one answer.*)

- Urban area, Stratum 1 or undefined
- Urban area, Stratum 2
- Urban area, Stratum 3
- Urban area, Stratum 4
- Urban area, Stratum 5
- Urban area, Stratum 6
- Rural area

Q13. Do you currently receive your own income (from work, business, pension, assistance, or any other source)? (*Choose one answer.*)

- Yes
- No

Q15. Approximately, what is your average monthly personal income? (Include salary, self-employment income, pension, rental income, assistance, or any other source. If your income varies, think of an average month.) (*Shown only if the respondent answered “Yes” to Question 13.*) (*Choose one answer.*)

- \$100 – \$500,000
- \$500,001 – \$1,000,000
- \$1,000,001 – \$1,500,000
- \$1,500,001 – \$2,500,000
- \$2,500,001 – \$4,000,000
- \$4,000,001 – \$7,000,000
- \$7,000,001 – \$10,000,000
- More than \$10,000,000
- Don’t know

Q17. Are you enrolled in a health social security entity (EPS, subsidized EPS)? (*Choose one answer.*)

- Yes, as a contributing member (*cotizante*)
- Yes, as a primary beneficiary

- Yes, but doesn't know whether as a contributing member or as a primary beneficiary
- Not enrolled
- Don't know

Q18. On average, how many times do you visit the doctor per year? (*Choose one answer.*)

- 0 times
- 1 time
- 2–3 times
- 4–6 times
- 7–10 times
- More than 10 times

Q19. How much do you pay or have deducted monthly to be covered by a health social security entity? (*Choose one answer.*)

- \$0
- \$100 – \$50,000
- \$50,001 – \$100,000
- \$100,001 – \$200,000
- \$200,001 – \$400,000
- More than \$400,000

Note: Take into account EPS, supplementary plans, prepaid health plans, policies, etc.

Q20. Approximately how much did you spend on healthcare services last month (excluding insurance premiums)? (*Choose one answer.*)

- \$0
- \$100 – \$30,000
- \$30,001 – \$80,000
- \$80,001 – \$150,000
- \$150,001 – \$300,000
- \$300,001 – \$600,000
- More than \$600,000
- Don't know

Question 21 asks respondents which of seven contraceptive brands/methods they recognize (Cyclofem, Depo Provera, Jadelle, Microgynon, Gynorelle, Yasminiq, Veroniq). For each brand a respondent indicates recognizing, the survey platform (LimeSurvey) repeats an identical five-item product-knowledge block (route of administration, prescription requirement, insurance coverage, ease of access, and awareness of comments), incrementing the underlying question numbers by nine each time. The Cyclofem instance of this block (Questions 22–26) is reproduced below.

Q22. For this product (Cyclofem), please indicate the route of administration that you know or believe to be correct: *(Choose one answer.)*

- Daily oral pill
- Weekly patch
- Monthly injection
- Quarterly injection (every 3 months)
- Subdermal implant
- Hormonal IUD
- Copper IUD
- Vaginal ring
- Female or male condom
- Emergency contraceptive pill
- Other (please specify)

Q23. Is a medical prescription required? *(Choose one answer.)*

- Yes
- No
- Don't know

Q24. Is it provided free of charge by your EPS or health insurer? *(Choose one answer.)*

- Yes
- No
- Don't know
- I do not have health insurance

Q25. How easy is it to obtain this product at each of the following places? *(Please select the appropriate answer for each row.)*

Scale: 1. Not obtainable / 2. Requires waiting days / 3. Requires waiting hours / 4. Immediately

	1	2	3	4	Don't know
EPS or health insurer					
Neighborhood pharmacy/drugstore					
Internet or social media					

Q26. Have you heard positive or negative comments about this product? (*Choose one answer.*)

- Only negative
- Mostly negative
- Generally neutral
- Mostly positive
- Only positive
- I have not heard any comments

e.g., positive: it might have health benefits

2.2 Description of Respondents

APPENDIX TABLE 2: Comparison between Survey and DHS Respondents

	Survey	DHS 2015
<i>Panel A. Age group</i>		
18 – 25	77.04	35.21
26 – 35	12.06	37.98
36 – 49	10.89	26.81
<i>Panel B. Educational attainment</i>		
No education	-	0.26
Primary	0.34	4.43
Secondary	27.03	29.37
Higher	72.64	65.94
<i>Panel C. Current contraceptive use</i>		
Yes	54.39	53.90
No	45.61	46.1
<i>Panel D. Contraceptive Method Conditional on Using Hormonal Contraceptives</i>		
Pills	40.00	25.97
Injections	6.00	38.96
LARC	51.00	34.89
Other	3.00	0.18

Notes: Entries are percentages of respondents within each column. The *DHS* column includes respondents from DHS 2015 aged 18 to 49 years who reported being in the contributory health insurance system. Panel D reports the distribution of contraceptive methods conditional on using a hormonal contraceptive.

2.3 Survey Evidence

APPENDIX TABLE 3: Survey Evidence of Contraceptive Knowledge

<i>Panel A. Number of contraceptive routes correctly identified</i>				
	Obs.	Mean	Std. Dev.	Mean as % of max score (%)
All sample (contributors)	89	1.07	0.79	15.25
High expenditure	76	1.12	0.78	15.98
Low expenditure	13	0.77	0.83	10.99
<i>Panel B. Awareness and route knowledge</i>				
	Obs.	Recognizing at least one brand (%)	Correctly identifying at least one route (%)	Correctly identifying no routes (%)
All sample (contributors)	89	87.64	75.28	24.72
High expenditure	76	88.16	78.95	21.05
Low expenditure	13	84.62	53.85	46.15

Notes: The table reports awareness of contraceptive brands and knowledge of their routes of administration for women responsible for financing their health insurance enrollment. Participants were asked about seven contraceptives; consequently, the count variable ranges from zero to seven. A route was classified as correct only when the participant both recognized the contraceptive brand and correctly identified its route of administration. Responses in which the brand was recognized but the route was incorrect or not provided were coded as zero and therefore did not increase the count. In Panel A, *Mean as % of maximum score* is calculated as the mean number of correctly identified routes divided by seven and multiplied by 100. Panel B reports the percentage recognizing at least one brand, correctly identifying at least one route, and correctly identifying no routes. *High expenditure* denotes women who have any spending in the last month on healthcare. *Low expenditure* denotes their counterparts.

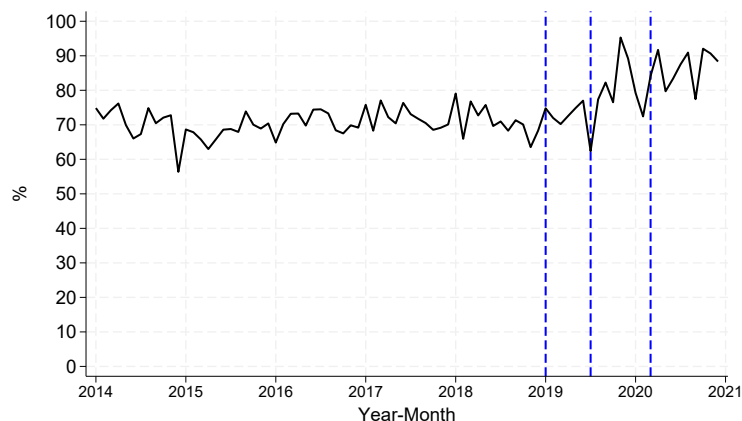
APPENDIX TABLE 4: Survey Evidence of Hassle Costs

<i>Panel A. Time required to obtain contraceptives</i>		
	Time required to obtain contraceptives	
	Insurance	OTC
Less than one hour	17.65	79.87
More than one hour	82.35	20.13
Observations	34	154
<i>Panel B. Travel time to the nearest location</i>		
	Travel time to the nearest location	
	Insurer delivery point	Retail pharmacy
Less than 10 minutes	5.41	77.60
Between 10 and 30 minutes	29.39	16.88
More than 30 minutes	65.20	5.52
Observations	296	308
<i>Panel C. Location where the most recent contraceptive was obtained</i>		
	Share (%)	Observations
Retail pharmacy or drugstore	61.35	154
Insurer dispensing facility	13.55	34
Other source	25.10	63
Applicable responses	100.00	251

Notes: Entries are percentages of respondents within each column. Panel A reports the time required to obtain the most recent contraceptive, including visits to the pharmacy, waiting in line, or scheduling an appointment with the appropriate physician. The Insurance column includes respondents whose most recent method was obtained at an insurer dispensing facility; the OTC column includes respondents whose most recent method was obtained at a retail pharmacy or drugstore. Panel B reports travel time from the respondent's home or workplace to the nearest insurer delivery point or retail pharmacy. Missing and *Does not know* responses are excluded separately for each location. Panel C reports the location where the respondent obtained her most recent contraceptive among women for whom the question was applicable. OTC denotes over-the-counter.

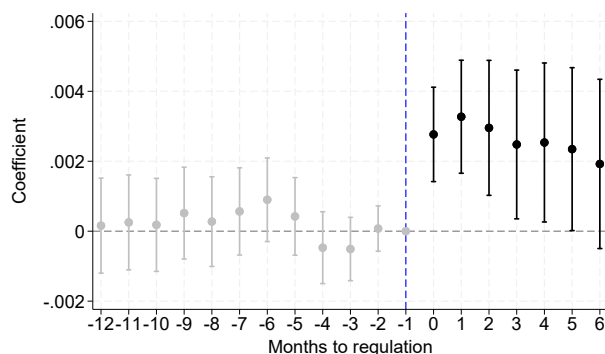
Appendix 3 Additional Descriptive Evidence

APPENDIX FIGURE 1: Share of Total Volume Through the OTC Segment



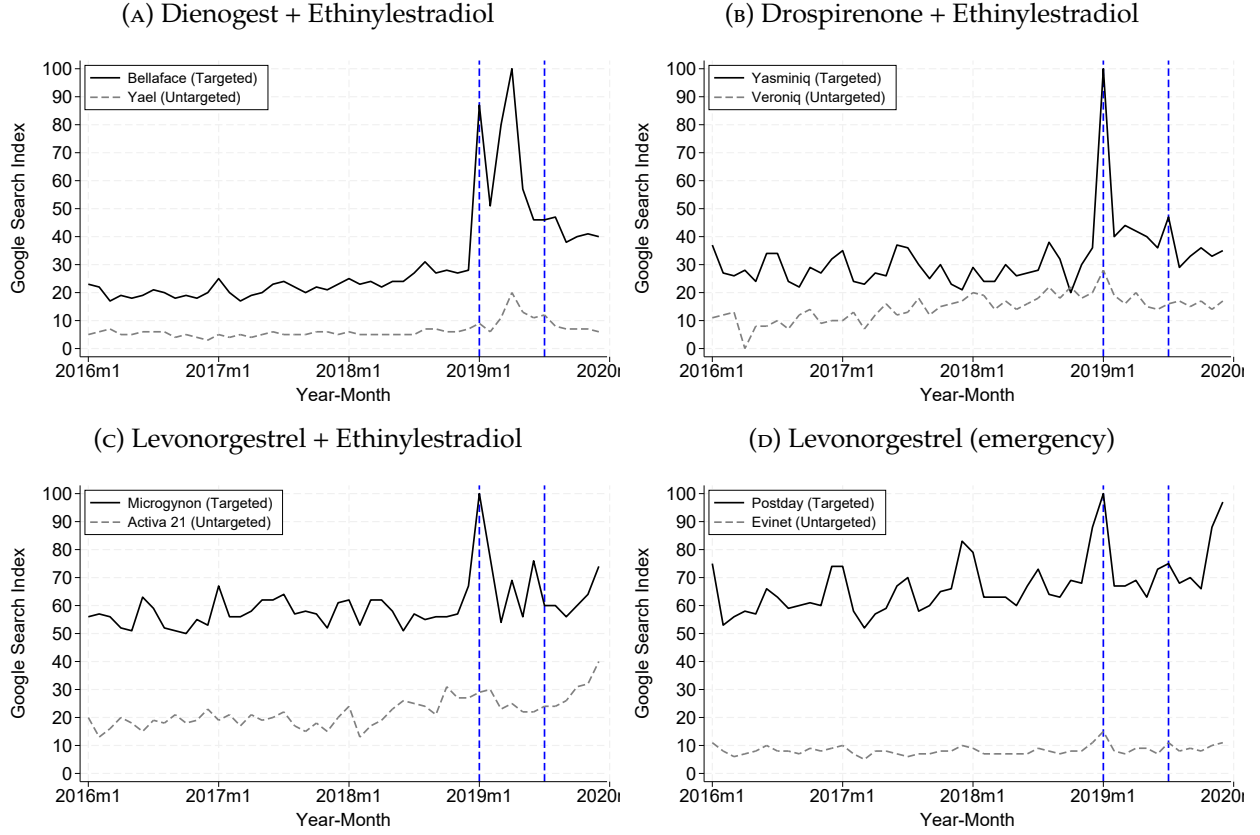
Note: Figure shows the share of the total volume of contraceptives distributed through the over-the-counter segment. The first blue line marks January 2019, when the first contraceptive price cap regulation (*Circular 07/2018*) came into effect. The second and third lines correspond to July 2019 (*Circular 09/2019*) and March 2020 (*Circular 10/2020*), marking the implementation of two additional contraceptive price cap regulations. Data comes from the Medication and Medical Device Price Information System (SISMED).

APPENDIX FIGURE 2: Comparison of Targeted Product Demand Among Informed and Uninformed Women



Note: Figure presents coefficients and 95% confidence intervals of an event study using as outcome an indicator for consuming targeted products. The treatment group are less-informed women or those with below-median pre-policy annual health spending (excluding prescriptions), the control group are their counterparts. The regression includes woman and month fixed effects; standard errors are clustered at the woman level. The dashed blue line represents the month when the price ceiling regulation went into effect.

APPENDIX FIGURE 3: Google Search Trends on a Subset of Contraceptives



Note: Figure presents the monthly Google Search Index for each pair of contraceptives by active ingredient from 2016-01 to 2019-12. The blue dashed lines correspond to the months when the contraceptive price ceilings were enacted. The light gray, long-dashed line represents untargeted contraceptives, while the solid black line represents targeted products.

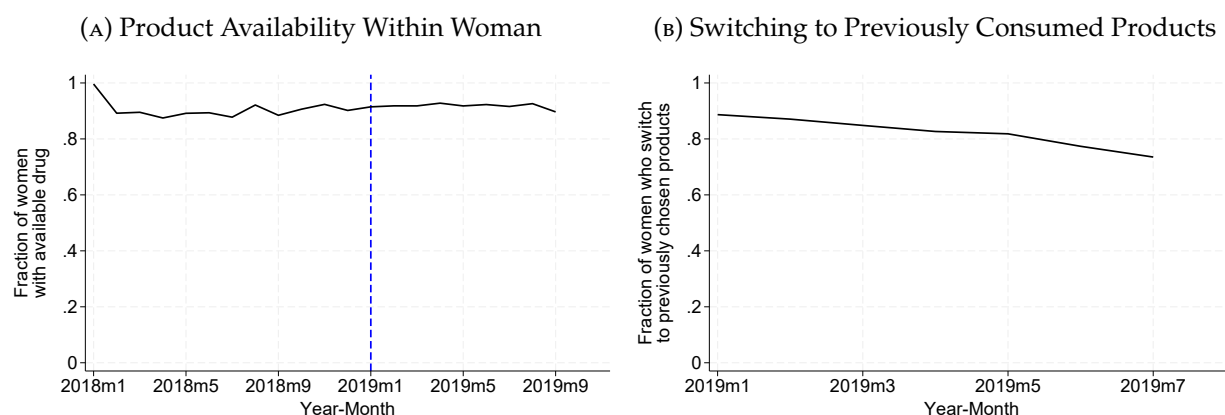
3.1 Competing Explanations in the Insurance Segment

This subsection explores additional competing explanations for the increase in demand for targeted products in the insurance data. First, insurers may have expanded their inventories of targeted products or discontinued untargeted ones, mechanically inducing women to switch after the regulation. To assess this possibility, Appendix Figure 4, Panel A plots, for each month t , the fraction of women whose contraceptive used in month $t - 1$ remained available through their insurer in month t . For approximately 90% of women, their incumbent product continued to be available after the policy, suggesting that forced switching due to product unavailability is an unlikely mechanism. Second, the price ceiling policy may have coincided with a period of high product experimentation among

women. However, Appendix Figure 4, Panel B shows that in January 2019, 90% of women were consuming contraceptives that they had used before 2018, suggesting that product experimentation is also an unlikely mechanism.

The pattern that women return to a previously consumed product after the regulation is also difficult to reconcile with consideration set probabilities. Under a consideration set model, women who were previously unaware of a product would be unlikely to switch to it after the regulation. Yet nearly 90% of switchers choose products they had used before, implying that the relevant barrier was not awareness but rather uncertainty about quality.

APPENDIX FIGURE 4: Evidence of Information Frictions in Contraceptive Use



Note: Panel A presents the fraction of women in month t whose $t - 1$ choice of contraceptive was offered by her insurer-municipality in month t . Panel B presents, among women who switch in month t , the fraction who switches towards products they had consumed before 2019m1. The dashed blue line presents the month when the first price ceiling regulation went into effect.

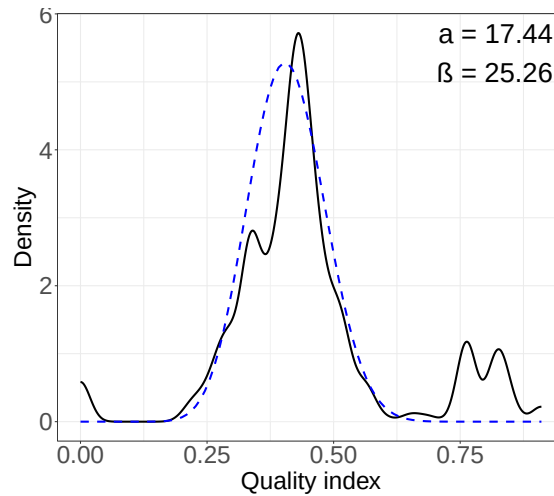
APPENDIX TABLE 5: Correlation between Wholesale Prices and Demand in INS Segment

	Market share	
	(1)	(2)
Log wholesale price	—	-0.004*** (0.0009)
Quality index	0.013 (0.017)	0.003 (0.016)
Route FE	Yes	Yes
Month FE	Yes	Yes
Observations	437	437
R ²	0.689	0.715

Note: Table presents OLS regressions of product market shares in the insurance segment on our measure of quality and the log of wholesale prices. Specifications include route of administration and month fixed effects. Market shares are constructed relative to the size of the potential market. Standard errors in parenthesis clustered at the product level.

3.2 Support for Model Assumptions

APPENDIX FIGURE 5: True Quality Distribution



Note: Figure presents the distribution of true quality in black and an estimated beta distribution in blue. The estimated parameters of the beta distribution are reported in the text box.

APPENDIX TABLE 6: Correlation between Regulation Status and Quality

Variable	$\mathbb{1}\{\text{Regulated}\}$	
	(1)	(2)
Quality index	0.437*** (0.072)	0.433*** (0.065)
Log price (OTC)		0.132*** (0.008)
<i>Fixed-effects</i>		
Route	Yes	Yes
Month	Yes	Yes
Observations	1,728	1,728
R ²	0.218	0.297

Note: Table presents OLS regressions of an indicator for whether the product is ever regulated on its quality measure in column (1) and additionally on its log effective price in column (2). Both specifications include route of administration and month fixed effects. Robust standard errors in parenthesis.

Appendix 4 Estimation Details

Data preparation. We append the product-month level data from IQVIA and from the insurance segment. Then, we restrict the estimation sample to the pre- and post-policy period from January 2018 to December 2019. We drop all months of products whose maximum inside market share over the sample period falls below 0.1% to avoid numerical instability in the contraction mapping and keep 87% of observations that constitute a balanced panel of products. Potential market size is set to 117% of observed monthly quantity in the reference period (December 2018), implying an outside option share of approximately 35%. We construct two demand-side instruments: a dummy variable for targeted products after the regulation in January 2019 and the predicted percentage change in prices for targeted products after the regulation. The [Gandhi and Houde \(2020\)](#) differentiation instruments with respect to the grams of estrogen and progestin are used as instruments on the supply side.

Demand estimation. We estimate the nonlinear parameters $\Phi^{nl} = (\alpha'_d, \rho_1, \gamma, \sigma)$ by minimizing the GMM objective using the BOBYQA algorithm implemented in the `nloptr` package in R ([Johnson, 2008](#)). We impose the constraints $\gamma \geq 0$ and $\sigma \in [0, 1]$.

To guard against local minima, we estimate the model from 10 starting values, drawn uniformly from $[0.1, 0.4]$ for α_1 , $[-0.4, 0]$ for α_2 , $[-0.4, 0]$ for α_3 , $[0, 10]$ for ρ_1 , $[0, 2]$ for γ , and $[0, 0.5]$ for σ . We use the following convergence tolerances (`ftol_rel` = `xtol_rel` = `xtol_abs` = 1^{-3} , and maximum 100 iterations). We select the solution with the lowest GMM objective and refine it with a smaller tolerance.

The contraction mapping tolerance is set to 1^{-12} throughout. Conditional on the nonlinear parameter estimates, we recover the linear parameters $\Phi^l = (\alpha_0, \tau, \rho_0)$ via 2SLS as described in equation (1). Estimation is parallelized across 12 cores.

Supply estimation. We estimate the supply-side nonlinear parameter $\tilde{\lambda}$ where $\lambda = \exp(\tilde{\lambda})$ to ensure $\lambda > 0$, by minimizing the supply-side GMM objective using BOBYQA, taking demand estimates as given. We initialize the algorithm at $\tilde{\lambda} = 0.1$ and use convergence tolerances of `ftol_rel` = 5^{-3} , `ftol_abs` = `xtol_rel` = `xtol_abs` = 1^{-3} with a

maximum of 5,000 iterations.

Conditional on $\hat{\lambda}$, we recover the linear supply parameters β^s via OLS on equation (4), and recover φ_t directly from the insurer's binding participation constraint.

Standard errors. We compute standard errors jointly for the demand and supply parameters using the sandwich GMM variance estimator, accounting for the two-step nature of the estimator. The standard error routine takes the estimated nonlinear demand and supply parameters as inputs and computes the full parameter covariance matrix in a single pass.

Appendix 5 Additional Model and Counterfactual Results

APPENDIX TABLE 7: Measures of Model Fit

Moment	Observed	Predicted
$\delta_{age,pills}$	0.281	0.278
$\delta_{age,inject}$	0.271	0.279
$\delta_{age,IUD}$	0.369	0.369
δ_{reg}	0.036	0.037
δ_{qual}	0.078	0.085
δ_{old}	-0.306	-0.307
δ_{young}	-0.350	-0.361
$\delta_{low\ income}$	-0.346	-0.353
$\delta_{high\ income}$	-0.339	-0.344
$\delta_{ins\ history}$	-0.347	-0.360
$\delta_{no\ ins\ history}$	-0.340	-0.341

Note: Table presents empirical moment from the health claims data and the model-predicted moment.

APPENDIX TABLE 8: Price Elasticity Matrix by Route of Administration in OTC Segment

Route	Conventional Pills	Emergency Pills	Implant	Injection	IUD
Conventional Pills	-1.722	0.027	0.001	0.031	0.023
Emergency Pills	0.010	-1.705	0.001	0.031	0.023
Implant	0.010	0.027	-1.731	0.031	0.022
Injection	0.010	0.027	0.001	-1.701	0.023
IUD	0.010	0.027	0.001	0.030	-3.652

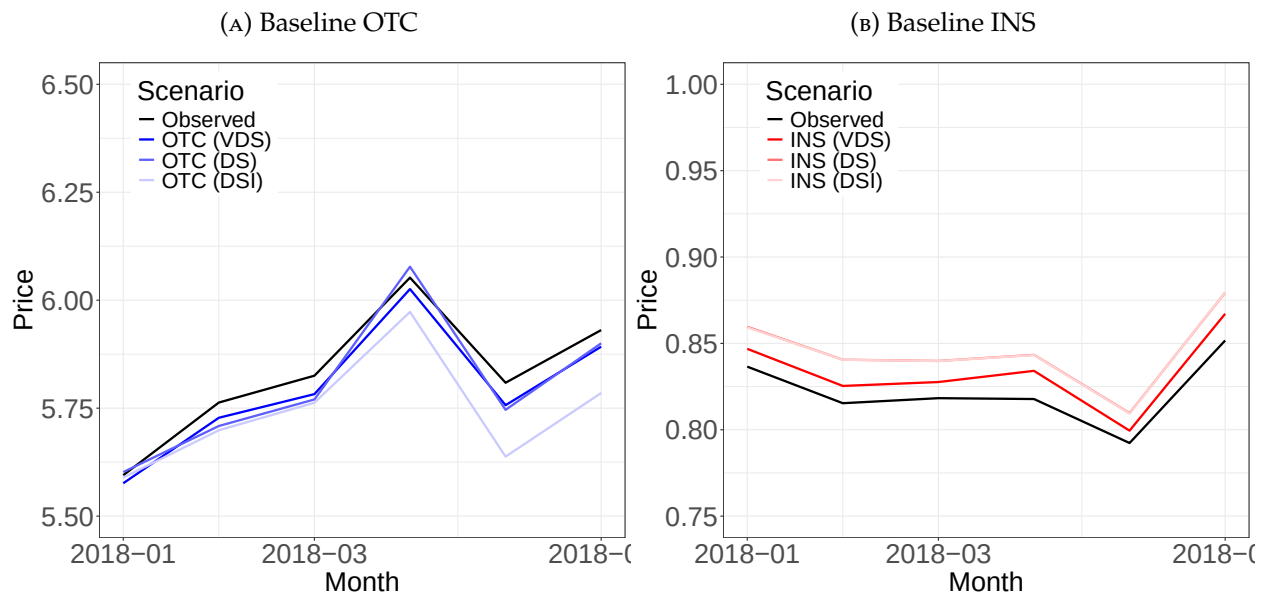
Note: Table presents the price elasticity matrix across routes of administration in the OTC segment during the pre-policy period. To obtain route-level elasticities from the product-level matrix, we aggregate by taking simple averages of own- and cross-price elasticities across all products within each route.

APPENDIX TABLE 9: Price Elasticity Matrix by Route of Administration in INS Segment

Route	Conventional Pills	Implant	Injection	IUD
Conventional Pills	-3.712	0.130	0.038	0.007
Implant	0.007	-3.590	0.038	0.006
Injection	0.007	0.130	-3.683	0.006
IUD	0.007	0.123	0.036	-3.700

Note: Table presents the price elasticity matrix across routes of administration in the INS segment during the pre-policy period. To obtain route-level elasticities from the product-level matrix, we aggregate by taking simple averages of own- and cross-price elasticities across all products within each route.

APPENDIX FIGURE 6: Counterfactual Price Trends



Note: Figure presents counterfactual and observed demand-weighted average per month during 2018. Panel A presents trends conditional on products offered OTC at baseline for the OTC-only counterfactual. Panel B presents price trends conditional on products that are offered through insurance at baseline in the INS-only counterfactual. Each panel presents price trends under four scenarios: one where product variety, demand, and supply are allowed to change; one where only demand and supply are allowed to change, and one where demand and supply are allowed to change and information frictions are removed.

APPENDIX TABLE 10: Diversion to Outside Option from Baseline OTC Products

Route	Observed (1)	DS (2)	DSI (3)
All types	0.391	0.275	0.275
Conventional Pill	0.392	0.275	0.276
Emergency Pill	0.389	0.274	0.274
IUD	0.399	0.279	0.279
Implant	0.379	0.265	0.265
Injection	0.390	0.275	0.275

Note: Table presents the demand-weighted diversion to the outside option conditional on products offered OTC at baseline. The diversion ratio is calculated as $\bar{D}_{0,t} = \sum_{j \in \mathcal{J}_{\text{OTC}}} \frac{s_{jt}}{\sum_{k \in \mathcal{J}_{\text{OTC}}} s_{kt}} \cdot \frac{\sum_i s_{i0t}}{\sum_i (1 - s_{ijt})}$. We report several scenarios: the observed scenario (column 1); the OTC-only counterfactual holding information status fixed (column 2); and the OTC-only counterfactual assuming full information (column 3). All counterfactual simulations use information from January to June 2018.

APPENDIX TABLE 11: Diversion to Outside Option from Baseline INS Products

Route	Observed (1)	DS (2)	DSI (3)
All types	0.394	0.683	0.684
Conventional Pill	0.399	0.698	0.698
IUD	0.402	0.703	0.702
Implant	0.387	0.665	0.666
Injection	0.395	0.690	0.690

Note: Table presents the demand-weighted diversion to the outside option conditional on product offered through INS at baseline. The diversion ratio is calculated as $\bar{D}_{0,t} = \sum_{j \in \mathcal{J}_{\text{INS}}} \frac{s_{jt}}{\sum_{k \in \mathcal{J}_{\text{INS}}} s_{kt}} \cdot \frac{\sum_i s_{i0t}}{\sum_i (1 - s_{ijt})}$. We report several scenarios: the observed scenario (column 1); the INS-only counterfactual holding information status fixed (column 2); and the INS-only counterfactual assuming full information (column 3). All counterfactual simulations use information from January to June 2018.